Forecasting U.S. Data Center Electricity Demand Growth in the Age of AI

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Abstract:

With the rapid expansion of artificial intelligence, cloud computing, and high-density data center infrastructure, electricity demand growth in the United States is entering a new phase. As a form of large-scale, continuously operating and highly concentrated electricity load, data centers are becoming one of the most important and most uncertain drivers of future U.S. electricity demand growth. However, substantial differences exist across institutional forecasts of both national electricity demand and data center electricity demand. Forecast ranges, time horizons, scenario assumptions, modeling approaches and analytical boundaries vary, making direct comparison and interpretation difficult.

This study examines forecasts of U.S. national electricity demand and U.S. data center electricity demand through a systematic multi-source comparative framework. Forecast data were collected and organized from government agencies, national laboratories, international organizations, consulting firms, financial research institutions and industry research organizations, including the U.S. Energy Information Administration (EIA), National Renewable Energy Laboratory (NREL), Lawrence Berkeley National Laboratory (LBNL), International Energy Agency (IEA), Electric Power Research Institute (EPRI), Princeton University, Low-Carbon Resources Initiative (LCRI), S&P Global, Boston Consulting Group (BCG), BloombergNEF, McKinsey & Company, and Rhodium Group. A structured MySQL forecast database was developed to standardize heterogeneous source data through data cleaning, unit conversion and scenario standardization. Separate forecast data systems were then constructed for national and data center electricity demand.

Methodologically, this study applies multi-source forecast trajectory comparison, benchmark-year forecast range analysis, pairwise difference heatmap analysis, and growth-intensity ranking to thoroughly evaluate forecast divergence, uncertainty and variation across institutions and scenarios. The results indicate that U.S. national electricity demand is likely to continue increasing gradually, with most forecasts falling around 4,300–5,100 TWh by 2030, 4,600–5,600 TWh by 2035, and 5,700–7,200 TWh by 2050, while more aggressive electrification or transition pathways may exceed 8,000 TWh. In contrast, U.S. data center electricity demand is projected to grow much faster and diverge earlier, with most 2030 forecasts concentrated around 400–650 TWh and high-growth AI scenarios approaching or exceeding 1,000 TWh. These findings suggest that data centers may become one of the fastest-growing and most uncertain sources of incremental U.S. load growth in the near future.

Overall, AI-driven data center expansion appears to be reshaping the traditional structure of U.S. electricity demand growth. Future U.S. electricity demand should therefore be interpreted not as a single deterministic trajectory, but as a structured range of possibilities shaped by different scenarios, institutional perspectives and technological pathways. The multi-source comparative framework and structured database developed in this study provide a foundation for future power system planning, grid investment, AI infrastructure development and policy analysis, while also supporting the continuous updating and long-term tracking of evolving data center electricity demand.

1. Introduction

1.1 Background

Before 2020, electricity demand in the United States remained relatively flat, with only moderate growth in national electricity load. However, recent evidence suggests that the United States is entering a new phase of more pronounced electricity demand growth. The U.S. Energy Information Administration (EIA) reported that U.S. net electricity generation reached a record high in 2025, totaling approximately 4,430 TWh, showing a 2.8% increase compared to 2024 [1]. EIA also projects that U.S. electricity use will continue to grow in 2026 and 2027, with increasing demand from computing data centers, which was identified as a major driver of this growth [2].

With the expansion of cloud computing and artificial intelligence, model training and inference increasingly require high-density computing infrastructure and supporting systems, as well as stable, large-scale electricity supply capable of meeting intensive and variable operational requirements. Data centers, which host these computing facilities, have therefore become one of the most important, urgent and uncertain drivers of U.S. electricity demand growth. Their scale, energy intensity and growth trajectory have changed significantly in recent years. According to the Lawrence Berkeley National Laboratory (LBNL), U.S. data centers consumed approximately 176 TWh of electricity in 2023, accounting for about 4.4% of total U.S. electricity consumption. By 2028, this figure could increase to approximately 325–580 TWh, implying that data center electricity demand may double or triple within only a few years [3], [4]. The International Energy Agency (IEA) also identifies AI as the most important driver of future global data center electricity demand growth. IEA projects that global data center electricity consumption could reach approximately 945 TWh by 2030, with the United States representing one of the largest contributors to this increase. In the United States, data centers may account for nearly half of total electricity demand growth by 2030 [5].

However, the exact magnitude and development pathway of this trend remain highly uncertain. Data center electricity demand is not a homogeneous variable. There are substantial differences across data center types and scales: traditional enterprise data centers, cloud computing campuses, and high-density facilities designed for AI training and inference differ in server configuration, rack power density, cooling systems, utilization rates and power reliability requirements. McKinsey notes that AI data centers are driving higher rack densities and more concentrated single-site power requirements, with some new campuses requiring electricity interconnection capacity ranging from 100 MW to nearly 1 GW [6]. At the same time, data centers are typically characterized by continuous operation, high load factors and large single-site capacity. If such new loads are concentrated in regions with limited distribution or transmission capacity, they may intensify pressure on local grids, power system reliability and cost. This means that data center electricity demand growth may reshape regional grid planning conditions through geographically concentrated, rapidly connected, and large-scale loads. Therefore, it is necessary to systematically organize forecasts from different institutions to compare forecast ranges, growth trajectories and divergence patterns. Such cross-source comparison provides a more reliable basis for evaluating the potential role of data center load in U.S. electricity demand growth and for informing energy system planning, grid investment and AI infrastructure development.

Recent developments in the electric utility sector further show that data center load is already directly affecting utility investment and planning decisions. Several U.S. electric utilities have adjusted capital investment plans, signed large-load agreements and planned new infrastructure in response to data

center growth. For example, American Electric Power increased its five-year capital investment plan to $78 billion [7], while Entergy also increased its four-year capital spending plan to $57 billion, driven largely by energy infrastructure needs associated with Meta's data center operations in Louisiana [8].

1.2 Research Motivation

The motivation for this study arises from a central challenge in current electricity demand forecasting: data center load is increasingly recognized as a major driver of future demand growth, but existing forecasts remain fragmented across institutions, scenarios and modeling frameworks. Government agencies, national laboratories, international organizations, consulting firms, market research groups, and financial institutions produce projections with different objectives, base years, time horizons, scenario assumptions and analytical boundaries. Without systematic standardization, these forecasts cannot be directly compared.

This challenge is especially significant for data center electricity demand. Unlike conventional aggregate load growth, data center demand is shaped by rapidly evolving technological and market conditions, including AI adoption, cloud computing expansion, chip efficiency, server utilization, cooling requirements, and grid interconnection constraints. Many data center forecasts also have shorter time horizons than national electricity demand forecasts, limiting their usefulness for long-term power-system planning.

A further motivation is that national demand and data center demand are often analyzed separately, even though they are directly connected. National forecasts provide the macro-level context for U.S. load growth, while data center forecasts identify the most important emerging sources of incremental demand. Comparing them within the same analytical framework helps evaluate whether data centers represent a structural driver that may alter assumptions for generation, transmission and regional grid planning.

1.3 Research Objectives

The study develops a structured comparative framework for analyzing U.S. electricity demand forecasts. The objective is to improve the interpretability of existing projections and examine how different assumptions shape alternative views of future U.S. electricity demand. To this end, this study:

- constructs a standardized multi-source forecast MySQL database
- organizes national demand and data center demand as separate but comparable forecast modules
- compares forecast trajectories, target-year ranges, pairwise differences and growth rankings
- interprets the implications of AI-driven data center electricity demand growth for system planning, grid investment, and energy policy

Overall, this study provides a systematic and interpretable comparison of U.S. electricity demand forecast uncertainty in the era of AI development. It argues that future electricity demand should be understood not as a single baseline trajectory, but as a structured range of possibilities shaped by institutional assumptions, technological change, and data center load growth.

2. Data Sources

This chapter introduces the data sources used in this study. The dataset was constructed from multiple federal agencies, national laboratories, international organizations, consulting firms, industry research groups and financial research institutions, providing a broad basis for comparing how different institutions project future U.S. electricity demand under the influence of AI-driven data center growth.

2.1 National Electricity Demand Data

For national electricity demand, this study includes projections from EIA, IEA, ICF, Citi Research, S&P Global, the National Renewable Energy Laboratory (NREL), BP, Princeton University, the Low-Carbon Resources Initiative (LCRI) and LBNL.

These sources represent different institutional perspectives on future U.S. electricity demand growth. EIA provides official reference and sensitivity cases; NREL and Princeton provide electrification and net-zero transition pathways; IEA, BP, S&P Global, Citi Research and ICF provide international, market-oriented or consulting-based outlooks; LCRI focuses on low-carbon transition demand envelopes; and LBNL provides implied national demand estimates linked to projected data center electricity consumption and electricity-demand shares.

a. EIA (published in 2025–2026) [9], [10]

EIA forecasts were collected from the *Annual Energy Outlook* (AEO) series. The dataset includes the AEO 2025 Reference Case and multiple AEO 2026 sensitivity cases covering different assumptions regarding economic growth, fuel supply, electricity demand, and zero-carbon technology cost. The forecasts span approximately from 2024 to 2050 and were reported in billion kilowatt-hours, which are numerically equivalent to TWh. The EIA dataset serves as the primary government reference framework in this study and provides both baseline and sensitivity trajectories for long-term U.S. electricity demand growth.

b. IEA (published in 2026) [11]

The IEA dataset was constructed as a derived U.S. electricity demand trajectory based on projected electricity-demand growth, U.S. population and per-capita electricity consumption assumptions. The series covers 2024 to 2030. Because IEA does not publish a standalone annual U.S. electricity demand forecast table for this dataset, the resulting trajectory should be interpreted as an IEA-style constructed demand pathway.

c. ICF (published in 2025) [12]

The ICF dataset represents a consulting-based national electricity demand outlook derived from an EIA baseline. ICF reports percentage growth assumptions, so the corresponding electricity demand values were derived from the reported growth rates. The dataset provides a high-growth consulting perspective on long-term U.S. electricity demand expansion.

d. Citi Research (published in 2025) [13]

The Citi Research dataset is based on a 2.8% compound annual growth rate (CAGR) outlook for U.S. electricity demand. Annual demand values from 2025 to 2040 were constructed by applying

the reported CAGR to an EIA baseline value. This trajectory represents a financial-research perspective on medium-term U.S. electricity demand growth.

e. S&P Global (published in 2025) [14]

The S&P Global dataset represents a market-oriented outlook for U.S. Lower 48 net on-grid electricity demand. The source reports benchmark-year values from 2024 to 2040. This dataset provides an industry and market-intelligence perspective on future electricity demand growth.

f. NREL (published in 2018) [15]

NREL forecasts were collected from the *Electrification Futures Study* (EFS), which is based on bottom-up load-profile modeling. The original dataset is organized by electrification level, technology advancement assumption, state, sector, subsector, hour and load in megawatts. Annual national electricity demand values were derived by aggregating hourly loads across all categories and converting the results into TWh. The dataset includes nine combinations of electrification level and technology advancement assumptions, preserving NREL's original scenario structure while allowing comparison within a standardized low–mid–high framework.

g. BP (published in 2025) [16]

The BP dataset was collected from *Energy Outlook 2025 – US Insights*. BP reports projected U.S. power generation values for selected benchmark years. Two scenarios are included: *Current Trajectory* and *Below 2°*. The latter reflects a more aggressive decarbonization pathway associated with stronger electrification and higher electricity demand.

h. Princeton (published in 2021) [17]

This dataset was collected from Princeton University's *Net-Zero America* project. The source provides long-term U.S. electricity demand projections under alternative decarbonization pathways from 2020 to 2050 at five-year intervals. The dataset includes low, mid and high transition pathways and represents a structured range of future electricity demand under different net-zero assumptions.

i. LCRI (published in 2024) [18]

The LCRI dataset was collected from *LCRI Net-Zero 2050* electricity demand trajectories. The values were digitized from published demand figures and cover benchmark years from 2020 to 2050. The dataset includes lower-bound, central and upper-bound trajectories reflecting different low-carbon transition assumptions.

j. LBNL (published in 2024) [3]

LBNL is included as a derived national electricity demand source rather than a direct national forecast provider. The *2024 United States Data Center Energy Usage Report* reports projected data center electricity demand together with data center shares of total U.S. electricity consumption. These relationships allow implied national electricity demand values to be inferred. The resulting trajectories should be interpreted as implied national electricity demand estimates derived from internally consistent data center demand and electricity-share assumptions.

2.2 Data Center Electricity Data

For U.S. data center electricity demand forecasts, this study includes projections and estimates from LBNL, the Electric Power Research Institute (EPRI), Boston Consulting Group (BCG), BloombergNEF, McKinsey & Company, IEA, Rhodium Group, S&P Global, and Jefferies. These sources were selected

because they directly estimate U.S. data center electricity demand or provide data that can be converted into comparable electricity-demand trajectories.

a. LBNL (published in 2024) [3]

The LBNL dataset was collected from the *2024 United States Data Center Energy Usage Report*. The report provides historical and projected U.S. data center electricity consumption through 2028, including electricity use from servers, storage, networking equipment, and supporting infrastructure. The dataset includes low-end and high-end future trajectories. Intermediate annual values were digitized from published figures.

b. EPRI (published in 2026) [19]

The EPRI dataset was collected from *Powering Intelligence: Updated U.S. Data Center Scenarios*. The source provides low, medium, and high scenarios for future U.S. data center electricity demand through 2030. The values were digitized from the updated solid-line scenarios reported in the study.

c. BCG (published in 2026) [20]

The BCG dataset was collected from *Solving the US Data Center Power Crunch*. The report provides annual U.S. data center electricity demand projections for hyperscale, colocation, and enterprise data centers, together with a higher-demand pathway. Both base and higher-demand trajectories are included in this study.

d. BloombergNEF (published in 2025) [21]

The BloombergNEF dataset was collected from *Power for AI: Easier Said Than Built*. Unlike most other sources, BloombergNEF reports data center power demand in GW, so it is converted into annual electricity demand using the relationship $TWh = GW \times 8.76$, assuming that the reported load represents average annual power demand.

e. McKinsey (published in 2024) [22]

The McKinsey dataset was collected from *Global Energy Perspective 2023*. The report provides a medium-scenario outlook for U.S. data center electricity consumption together with projected data center shares of total U.S. power demand. Electricity-demand trajectories were digitized from the published figures, while the reported electricity-demand shares were preserved separately for later comparison.

f. IEA (published in 2025) [23]

The IEA dataset was collected from the chart *Data Centre Electricity Consumption by Region, Base Case, 2020–2030*. Only the United States component of the regional dataset was extracted.

The original values were already reported in TWh and therefore required no unit conversion.

g. Rhodium (published in 2026) [24]

The Rhodium Group dataset was collected from *The Impacts of Rising Electricity Demand from Data Centers on US Energy and Emissions*. The source provides baseline and high-growth data center pathways linked to energy-system and emissions modeling.

h. S&P Global (published in 2024) [25]

The S&P Global dataset was collected from *US Datacenter and Energy Outlook – Powering the AI Economy*. The source provides near-term U.S. data center electricity demand projections supported by benchmark-year values and comparison figures. Intermediate annual values were digitized from the published trajectories.

i. Jefferies (published in 2024) [26], [27]

The Jefferies dataset was collected from *How Data Centers Are Shaping the Future of Energy Consumption*, which summarizes Jefferies Equity Research analysis on AI infrastructure and electricity demand growth. Because the complete underlying equity research report is not publicly available, the values used in this study were digitized from the forecast comparison chart in a World Resources Institute article [27], which includes the Jefferies projection.

2.3 Data Scope and Source Heterogeneity

The sources included in this study differ substantially in forecast scope, reporting format, time horizon, and analytical boundary. National electricity demand forecasts generally represent broader economy-wide or power-system pathways and often extend up to 2050, while data center forecasts focus on shorter horizons like 2030, reflecting the rapid evolution of AI infrastructure deployment, data center construction, and grid interconnection.

The sources also differ in reporting structure. Some provide complete annual electricity demand tables, while others report benchmark-year values, growth rates, scenario ranges, power-load estimates, or electricity-demand shares that require additional processing before comparison. Scenario structures likewise vary across institutions, including reference cases, electrification pathways, net-zero transitions, market outlooks, and AI-driven demand-growth assumptions.

These forecasts therefore do not represent a unified dataset generated under a single modeling framework. Instead, they form a heterogeneous collection of institutional perspectives on future U.S. electricity demand. This heterogeneity is central to the research design because it allows the study to evaluate forecast uncertainty, divergence and institutional differences across multiple analytical frameworks.

To enable systematic comparison, the collected forecasts were standardized in terms of unit, year, forecast category, scenario classification and data format. The following chapter describes how these heterogeneous source materials were converted into a structured database for subsequent visualization and comparative analysis.

3. Data Processing

This chapter describes the workflow to convert heterogeneous source materials into a structured and analysis-ready forecast database, including source-specific data collection, database design, manual transcription or digitization, data standardization, querying, and quality checks.

3.1 Data Collection

The data collection process aimed to produce a multi-source forecast dataset covering both U.S. national electricity demand and U.S. data center electricity demand. Accordingly, the data collection process emphasized three priorities: identifying publicly available and representative forecast sources; preserving the original scenario structure of each source; and organizing the collected values into a format suitable for subsequent standardization and database integration. The two categories were processed separately because they differ in scale, format, forecast horizon and analytical focus.

The data were collected from institutional reports, research articles, consulting reports, market outlooks, official datasets and related materials. In many cases, complete annual numerical tables were not publicly available. Instead, forecast information was presented through figures, dashboards, benchmark-year values, or textual descriptions. As a result, different extraction methods were applied depending on the reporting format of each source. When annual tabular data were available, the values were directly extracted. When benchmark values were explicitly labeled in figures or dashboards, they were manually recorded. When forecasts were presented only as graphical trajectories, data points were extracted through high-resolution digitization. When sources reported only growth rates, power-load values, data center electricity shares, or per-capita electricity-use assumptions, the annual demand values were derived through explicit calculations.

For example, EIA forecasts were obtained directly from the official Annual Energy Outlook tables and entered the database as tabular values. NREL forecasts were derived from the Electrification Futures Study hourly load dataset by aggregating across states, sectors, subsectors and hours before conversion into annual values. EPRI forecasts were extracted from published forecast figures because complete annual numerical tables were not available. The IEA national electricity demand series was constructed from population, per-capita electricity consumption and annual growth assumptions. BloombergNEF reported data center power demand in GW, which was converted into annual electricity consumption in TWh. LBNL's data center forecasts were used directly for the data center database, while the reported data center share of total U.S. electricity consumption was used to derive implied national electricity demand values.

Throughout the collection process, the original source, scenario name, year range, reporting unit, and acquisition method were recorded for each forecast curve. Preserving this information improves traceability and distinguishes original reported values from processed values.

3.2 Database Design

After data collection, this study constructed *us_dc_energy_forecast*, a structured MySQL database designed to store, organize and compare U.S. electricity demand forecasts. The database was organized into two separate but parallel modules: national electricity demand forecasts and data center electricity demand forecasts.

The database contains four core tables: *forecast_metadata*, *forecast_values*, *dc_forecast_metadata*, and *dc_forecast_values*. The first two tables store U.S. national electricity demand forecasts, while *the second two store* U.S. data center electricity demand forecasts.

The database follows a metadata–value structure. Each forecast curve is first defined in a metadata table, recording the source institution, original scenario name, standardized scenario classification, regional scope, year range, original unit and some methodological notes. The corresponding annual numerical values are then stored in the associated value table. One metadata record may correspond to multiple annual value records, forming a one-to-many relationship. This structure preserves the original context of each forecast curve while supporting flexible querying for visualization and analysis.

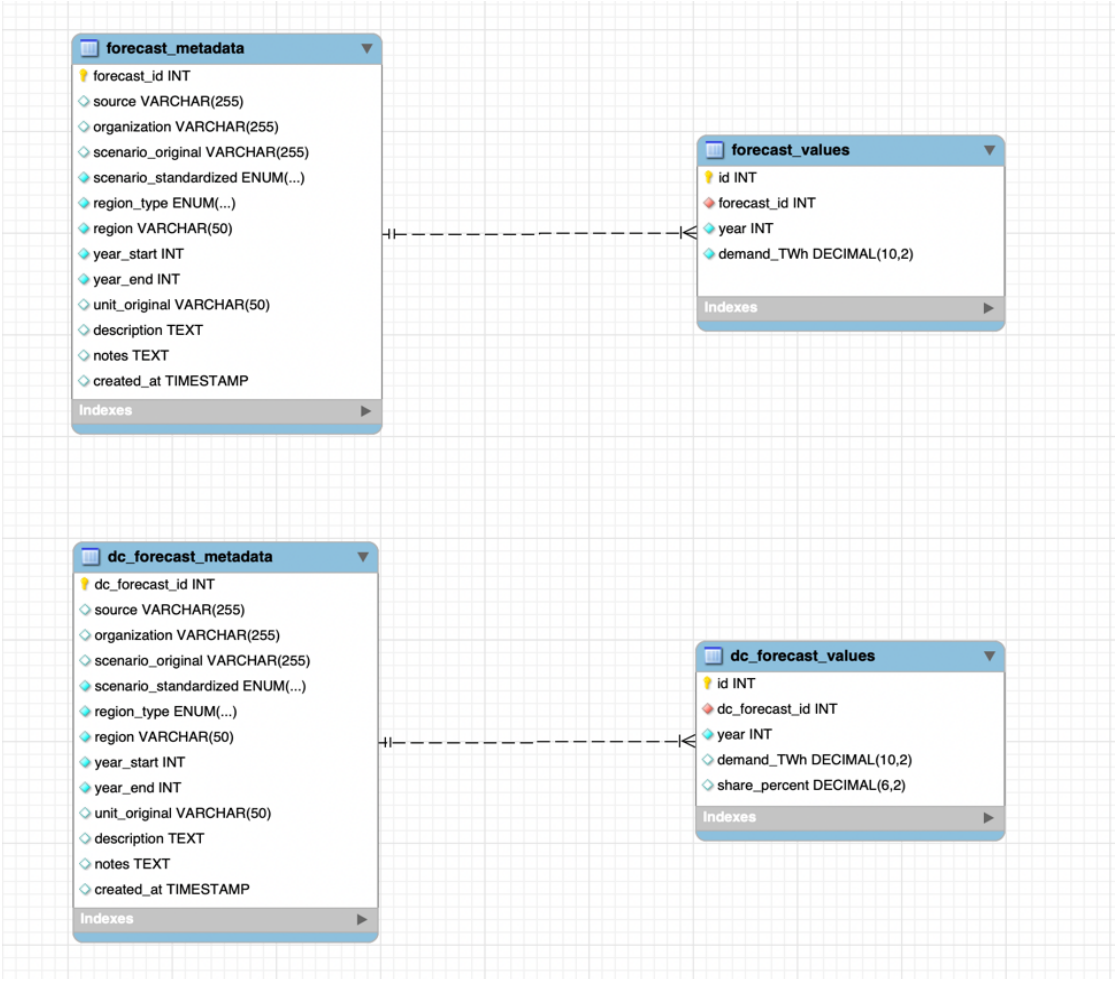


Figure 3.1 Database ER Diagram

As shown in the ER diagram (Figure 3.1), *forecast_metadata.forecast_id* serves as the primary key in the national electricity demand module, while *forecast_values.forecast_id* serves as the foreign key. Similarly, *dc_forecast_metadata.dc_forecast_id* serves as the primary key in the data center electricity demand module, and *dc_forecast_values.dc_forecast_id* serves as the foreign key. Each value record must correspond to an existing metadata record. Both value tables include a uniqueness constraint on *(forecast_id, year)* or *(dc_forecast_id, year)* to prevent duplicate annual entries within the same forecast curve.

Table 3.1 MySQL forecast_metadata Schema

Field	Data Type	Description
forecast_id	INT AUTO_INCREMENT PRIMARY KEY	Unique identifier for each curve
source	VARCHAR(255)	Short name of the data source
organization	VARCHAR(255)	Full name of the source institution
scenario_original	VARCHAR(255)	Original scenario name
scenario_standardized	ENUM('low','mid','high')	Standardized scenario category
region_type	ENUM('national','state','ISO')	Geographic level of the forecast
region	VARCHAR(50)	Geographic region (USA)
year_start	INT	Start year of the forecast curve
year_end	INT	End year of the forecast curve
unit_original	VARCHAR(50)	Original reported unit
description	TEXT	Brief description of the forecast
notes	TEXT	Methodological and processing notes
created_at	TIMESTAMP	Timestamp when created

The *forecast_metadata* table (Table 3.1) stores contextual information for each national electricity demand forecast curve, meaning that each scenario is represented as a separate metadata record.

Table 3.2 MySQL forecast_values Schema

Field	Data Type	Description
id	INT AUTO_INCREMENT PRIMARY KEY	Unique identifier for each record
forecast_id	INT	Forecast curve identifier
year	INT CHECK(year BETWEEN 2000 AND 2100)	Forecast year
demand_TWh	DECIMAL(10,2)	Standardized value in TWh

The *forecast_values* table (Table 3.2) stores annual national electricity demand values standardized to TWh. Through the *forecast_id* foreign key, each annual value can be traced to its corresponding source and scenario. Each forecast curve therefore contains multiple annual demand records through its one-to-many relationship with *forecast_metadata*.

Table 3.3 MySQL dc_forecast_metadata Schema

Field	Data Type	Description
dc_forecast_id	INT AUTO_INCREMENT PRIMARY KEY	Unique identifier for each curve
source	VARCHAR(255)	Short name of the data source
organization	VARCHAR(255)	Full name of the institution
scenario_original	VARCHAR(255)	Original scenario name
scenario_standardized	ENUM('low','mid','high')	Standardized scenario category
region_type	ENUM('national','state','ISO')	Geographic level of the forecast
region	VARCHAR(50)	Geographic region (USA)
year_start	INT	Start year of the forecast curve

Field	Data Type	Description
year_end	INT	End year of the forecast curve
unit_original	VARCHAR(50)	Original reported unit
description	TEXT	Brief description of the forecast
notes	TEXT	Methodological and processing notes
created_at	TIMESTAMP	Timestamp when created

The *dc_forecast_metadata* table (Table 3.3) has a structure similar to *forecast_metadata*, but is specifically designed for data center demand. The table preserves original scenario names and processing notes, allowing the analysis to distinguish among directly reported, digitized, converted and derived values.

Table 3.4 MySQL dc_forecast_values Schema

Field	Data Type	Description
id	INT AUTO_INCREMENT PRIMARY KEY	Unique identifier for each record
dc_forecast_id	INT	Forecast curve identifier
year	INT CHECK(year BETWEEN 2000 AND 2100)	Forecast year
demand_TWh	DECIMAL(10,2)	Standardized value in TWh
share_percent	DECIMAL(6,2)	Percentage of data center demand in total demand

The *dc_forecast_values* table (Table 3.4) stores annual data center electricity demand values. Unlike the national electricity demand value table, it includes an additional *share_percent* field representing the share of total U.S. electricity consumption attributed to data centers. This field is particularly relevant for sources such as LBNL and McKinsey, which report both data center electricity demand and its share of total U.S. electricity use. It also supports analysis of the relationship between data center load growth and total national electricity demand.

The design of this database provides three advantages. First, the metadata–value structure separates contextual forecast information from annual numerical values while preserving source traceability. Second, national electricity demand and data center electricity demand are organized into parallel table structures, avoiding analytical ambiguity between datasets with different scales and meanings. Third, standardized fields such as *scenario_standardized*, *region_type*, *year*, and *demand_TWh* support comparison across institutions, scenarios and forecast periods, forming the foundation for subsequent SQL querying, visualization and forecast divergence analysis.

3.3 Data Standardization

The next step was to standardize the raw data so that all forecast curves could be stored using a consistent database structure.

a. Analytical Classification

Each forecast curve was categorized as either a U.S. national electricity demand forecast or a U.S. data center electricity demand forecast. National electricity demand forecasts were entered into

the *forecast_metadata* and *forecast_values* tables, while data center electricity demand forecasts were entered into the *dc_forecast_metadata* and *dc_forecast_values* tables.

b. Unit Standardization

The database uses terawatt-hours (TWh) as the standard unit for electricity demand. Values originally reported in any other units are converted into TWh in the value table to simplify analysis.

c. Scenario Standardization

The standardized categories used in this study are *low*, *mid*, and *high*, regardless of their original scenario names. When a source provides a clear scenario range, the lower trajectory is classified as *low*, the central or reference trajectory as *mid*, and the upper trajectory as *high*. Single-pathway forecasts are generally classified as *mid* unless explicitly presented as upper-bound or high-demand projections.

d. Temporal Standardization

The database records all observations by calendar year. For sources providing annual values, the data were entered directly by year. For sources reporting only benchmark years, such as 2030, 2040, or 2050, only the reported years were entered. For graphical trajectories, annual values were extracted for years that could be clearly identified from the original figures. The *year_start* and *year_end* fields record the temporal coverage of each forecast curve, while the value tables store the available annual observations.

After standardization, each forecast curve was written into an independent SQL insertion file or scenario-specific SQL file. Each file follows the same general structure: a metadata record is first inserted to define the source, scenario, region, year range, unit, and processing notes, followed by insertion of the corresponding annual values into the associated value table. This process was repeated for each institution and scenario curve.

3.4 Data Querying

After that, two core query files were used to extract the datasets required for subsequent analysis.

```sql
-- ============================================================
-- forecast_query.sql
-- Purpose:
-- Query total U.S. electricity demand forecasts only.
-- ============================================================

SELECT
    v.year,
    v.demand_TWh,
    m.source,
    m.scenario_original,
    CONCAT(m.source, ' | ', m.scenario_original) AS curve

FROM forecast_values v
JOIN forecast_metadata m
    ON v.forecast_id = m.forecast_id

ORDER BY
    curve,
    v.year;
```

```sql
-- ============================================================
-- dc_forecast_query.sql
-- Purpose:
-- Query U.S. data center electricity demand forecasts only.
-- ============================================================

SELECT
    v.year,
    v.demand_TWh,
    v.share_percent,
    m.source,
    m.scenario_original,
    CONCAT(m.source, ' | ', m.scenario_original) AS curve

FROM dc_forecast_values v
JOIN dc_forecast_metadata m
    ON v.dc_forecast_id = m.dc_forecast_id

ORDER BY
    curve,
    v.year;
```

Figure 3.2 Two Forecast Queries

The *forecast_query.sql* file (Figure 3.2 Left) extracts U.S. national electricity demand forecasts, while *dc_forecast_query.sql* (Figure 3.2 Right) extracts U.S. data center electricity demand forecasts. Both

queries combine annual electricity demand values with their corresponding source and scenario information to produce structured datasets for analysis.

For U.S. national electricity demand forecasts, *forecast_query.sql* joins the *forecast_values* and *forecast_metadata* tables using *forecast_id*. The query output includes forecast year, annual electricity demand, data source, original scenario name and a *curve* field generated by combining *source* and *scenario_original*. The *curve* field uniquely identifies each forecast trajectory. The query output follows a long-format structure in which each row represents one annual electricity demand value associated with a specific source, scenario and year. Using *ORDER BY curve, v.year* arranges each forecast trajectory chronologically for subsequent processing and visualization.

For U.S. data center electricity demand forecasts, *dc_forecast_query.sql* applies the same general logic by joining *dc_forecast_values* and *dc_forecast_metadata* through *dc_forecast_id*, retaining the *share_percent* field for comparable analysis. This query also produces a long-format dataset ordered by curve and year. Each row corresponds to one annual data center electricity demand value for a specific source, scenario and year.

3.5 Data Quality and Consistency Checks

This section focuses on quality control and consistency checks during database construction and data processing. These checks were used to reduce errors introduced by manual transcription, digitization, unit conversion, derived calculations and SQL insertion.

a. Schema-level Checks

   The database schema provides the first level of quality control. Metadata and value tables are linked through primary and foreign keys, ensuring that each annual value can be traced to a specific forecast source and scenario. The value tables also include constraints preventing duplicate annual entries within the same forecast curve. In addition, the *scenario_standardized* field is restricted to *low*, *mid*, and *high* to maintain consistency in scenario classification.

b. Curve-level Checks

   Each forecast curve was also checked individually. The reviewed items included *source, scenario_original, scenario_standardized, year_start,* and *year_end,* together with the minimum year, maximum year, and total number of annual records stored in the associated value table. These checks verified consistency between metadata records and annual observations while identifying missing years, duplicate years, or incorrectly linked forecast identifiers.

c. Unit Conversion and Derivation Checks

   Key unit conversions and derived calculations were checked for consistency and numerical reasonableness. Because all forecast values were standardized to TWh, conversion logic had to remain consistent across different reporting units.

d. Magnitude and Trend Checks

   Magnitude and trend checks were also performed. National electricity demand forecasts are generally expected to remain on the order of several thousand TWh, while data center electricity demand forecasts are typically several hundred TWh.

Overall, these procedures improved the reliability, traceability and internal consistency of the compiled database and established a consistent foundation for subsequent visualization, comparison, and forecast divergence analysis.

4. Methodology

Building on the processed data, this chapter describes the analytical methods used to compare institutional electricity demand forecasts. Because the objective of this study is to establish a comparative framework for interpreting differences across existing forecasts, the methodology focuses primarily on comparative visualization and forecast divergence analysis.

4.1 Cross-Source Forecast Comparison

This study first applies cross-source forecast comparison as the foundational analytical method. Multi-source forecast trajectory plots are used to place forecast curves from different institutions within the same coordinate system, allowing direct comparison of how different institutions project future electricity demand growth for the same analytical object.

The national electricity demand plot covers approximately 2020 to 2050 and examines long-term changes, while the data center electricity demand plot focuses on 2022 to 2035 and examines near- and medium-term uncertainty associated with AI-driven data center expansion.

Each forecast curve is uniquely identified by combining *source* and *scenario_original* into a *curve* label. This labeling structure preserves both institutional identity and original scenario information, allowing each plotted trajectory to be traced directly to its corresponding database record. The Python scripts read the queried long-format tables from MySQL, convert *year* into integer format and *demand_TWh* into floating-point format, then sort the data by curve and year before plotting each forecast trajectory separately.

Full forecast trajectory plots provide a direct visualization of the overall forecast structure, showing differences in forecast horizon, revealing differences in projected growth intensity, and helping to identify when divergence begins to widen. They also support data validation and interpretation. Because all curves are generated from the same standardized database, abnormal discontinuities, incorrect magnitudes, unit-conversion errors, or year misalignments can often be identified visually.

4.2 Forecast Range Analysis

This study next applies forecast range analysis to compare institutional forecasts at selected benchmark years. Forecast range analysis extracts forecast values for a specific target year and summarizes the available scenarios for each institution as an interval consisting of a lower bound, a central base value and an upper bound. This method provides a clearer cross-sectional comparison of forecast levels and uncertainty ranges at the same point in time.

For U.S. national electricity demand, 2030, 2035 and 2050 are selected as benchmark years, representing near-term planning, intermediate transition and long-term horizons, respectively. For U.S. data center electricity demand, 2028 and 2030 are selected because most data-center-specific forecasts focus on near-term and medium-term development.

The range plots are constructed using a consistent procedure. First, all forecast curves are retrieved from the database, and only curves covering the target year are retained. Each curve is then classified according to its *scenario_standardized* and *scenario_original* fields. Lower-demand scenarios are assigned to the *low* category, higher-demand scenarios to the *high* category, and central trajectories include *mid*, *medium*, *reference*, *baseline*, *projected* and *central* cases. For each institution, the low and high values define the forecast interval, while the mid value (or, when no mid case exists, the

midpoint of the low and high values) is displayed as the central marker. Institutions providing only a single forecast curve are treated as having a mid-case estimate.

The visual structure of the range plots distinguishes central forecast levels from internal institutional uncertainty. The vertical interval represents the low–high forecast range for each institution, while the square marker indicates the base or projected forecast. This structure allows simultaneous comparison of two dimensions: the position of each institution's central forecast and the width of its internal scenario range. Methodologically, forecast range analysis provides a clearer basis for subsequent comparison of uncertainty across institutions, forecast horizons, and demand categories.

4.3 Difference Heatmap Analysis

This study applies difference heatmap analysis to quantify pairwise forecast differences across institutions and scenarios. The purpose of the heatmap is to calculate forecast differences directly for the same target year, providing a clearer representation of the magnitude and direction of forecast divergence.

The benchmark years for difference heatmap analysis are consistent with the forecast range analysis, allowing the two methods to complement each other. The range plots summarize forecast intervals for each institution, while the heatmaps quantify pairwise differences across forecasts.

The basic calculation is:

$$Difference_{(i,j)} = Forecast_i - Forecast_j$$

where $Forecast_i$ represents the forecast value of the row institution or scenario in the target year, and $Forecast_j$ represents the forecast value of the column institution or scenario in the same year. Each cell therefore represents the row forecast minus the column forecast measured in TWh. Positive values indicate that the row forecast is higher than the column forecast, while negative values indicate the opposite. Diagonal cells compare a forecast with itself and therefore contain no meaningful difference.

The heatmaps are constructed at two analytical levels. The first is the institution-level central difference heatmap, which compares representative central forecasts across institutions. The second is the within-institution scenario difference heatmap, which compares scenario-level differences within the same institution. The second type is generated only when an institution provides two or more forecast curves covering the target year.

The heatmap addresses the limitation of full trajectories and range analysis by organizing all pairwise differences into a matrix structure, providing a compact and systematic representation of forecast divergence, particularly useful for identifying institutions with similar projections, institutions that deviate substantially from others and changes in forecast divergence over longer forecast horizons.

4.4 Forecast Level and Growth Intensity Ranking

Finally, this study applies forecast level and growth intensity ranking to compare institutional forecasts in a ranked format. This method converts forecast results into direct comparative indicators, allowing comparison of both target-year forecast levels and the growth intensity implied by different projections.

The ranking analysis consists of two components. The first is target-year forecast level ranking, which compares the representative forecast value of each institution in a selected benchmark year. This indicator evaluates the absolute forecast level in a given year. The second is growth intensity ranking,

which compares the magnitude of growth from each institution's forecast starting point to the target year. Two metrics are used: growth multiple and Compound Annual Growth Rate (CAGR).

For target-year forecast level ranking, the benchmark years are still consistent with the forecast range analysis. Only one representative forecast value is selected for each institution in each target year. When an institution provides only one forecast curve, that curve is treated as the representative mid-case estimate. When two curves are available, their average is used as the representative value. When three or more curves are available, the central scenario is selected; where several curves qualify as central, as with the three medium-electrification curves of NREL EFS, their average is used. For this reason, the NREL values in the ranking tables differ slightly from the NREL central values shown in the range plots and heatmaps. This procedure ensures that the ranking reflects institution-level representative forecasts rather than the number of scenarios provided by each institution.

The target-year forecast ranking indicator is:

$$D_{(i,t)}$$

where $D_{(i,t)}$ indicates the representative electricity demand forecast of institution i in target year t measured in TWh. For each benchmark year, institutions are ranked according to $D_{(i,t)}$. Higher values indicate more aggressive electricity demand projections, while lower values indicate relatively more conservative forecasts.

The growth intensity ranking further evaluates the magnitude of growth from each institution's forecast starting year to the target year. The first metric is growth multiple:

$$Growth\ Multiple_{(i,t)} = \frac{D_{(i,t)}}{D_{(i,t_0)}}$$

where $D_{(i,t_0)}$ denotes the representative forecast value in the first available year t_0, and $D_{(i,t)}$ indicates the representative value in target year t. The growth multiple measures how many times larger the target-year electricity demand is relative to the starting-year value.

The second metric is CAGR:

$$CAGR_{(i,t)} = \left(\frac{D_{(i,t)}}{D_{(i,t_0)}}\right)^{\frac{1}{t-t_0}} - 1$$

which converts forecast growth across different time horizons into an annualized growth rate, making cross-institution comparison more consistent than total growth multiple alone. Institutions are ranked according to CAGR for each benchmark year. Higher CAGR values indicate stronger projected growth intensity, while lower values indicate more moderate growth pathways.

This method is essential for data center electricity demand forecasts. Data center projections differ not only in target-year electricity demand levels, but also in implied growth intensity. Some institutions project very high target-year demand values without implying the fastest growth rates, while others imply rapid growth because their starting values are relatively low. Combining target-year forecast ranking with CAGR ranking therefore distinguishes between the absolute scale of projected demand and the relative intensity of projected growth.

5. Data Analysis

This chapter presents the analysis of U.S. national electricity demand forecasts and U.S. data center electricity demand forecasts based on the database constructed in the previous chapters.

5.1 Cross-Source Forecast Trajectory Analysis

Cross-source forecast trajectory analysis places all forecasts in a single plot to show the overall trend in projected electricity demand. Each institution's curves are also plotted individually in Appendix I (national demand) and Appendix II (data center demand).

5.1.1 National Electricity Demand Forecast Trajectories

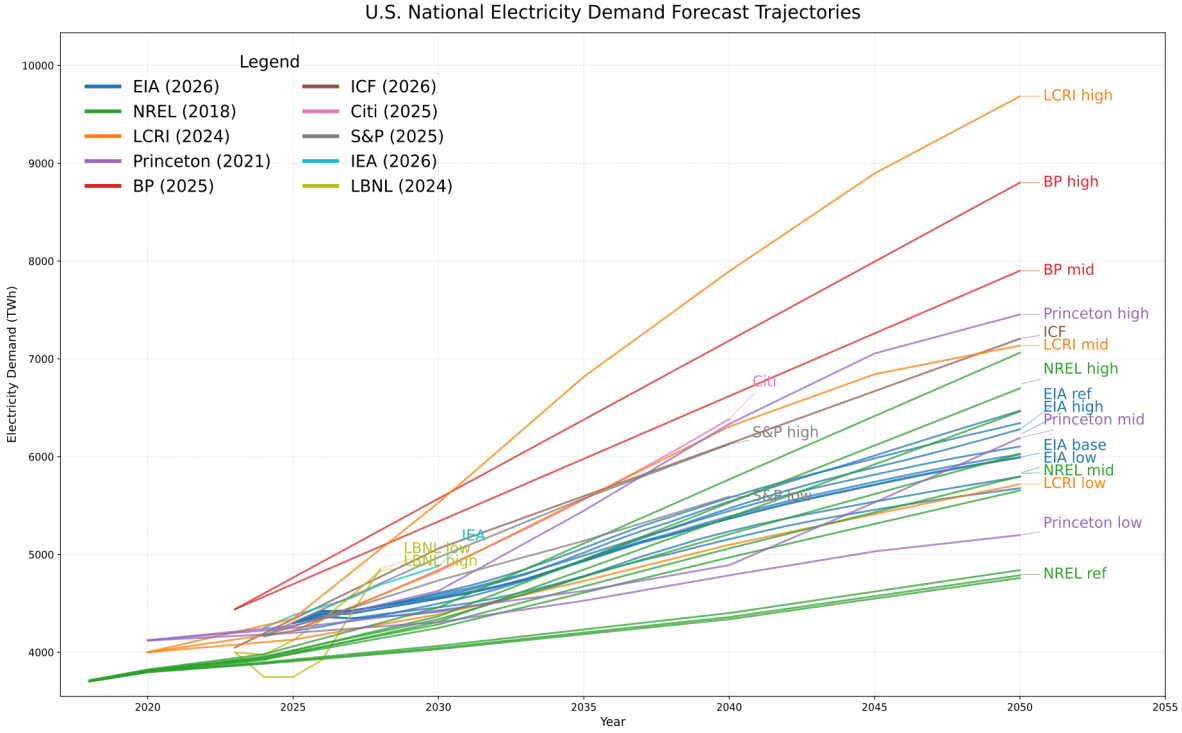


Figure 5.1.1 National Electricity Demand Forecast Full Trajectories

Figure 5.1.1 presents multi-institutional forecast trajectories for U.S. national electricity demand. It is clear that U.S. electricity demand will continue to grow over the coming decades. The forecast envelope formed by multiple institutional projections shows that near-term and medium-term forecasts are relatively concentrated, while long-term forecasts gradually diverge.

From a temporal perspective, between 2020 and 2030, differences across institutional forecasts are already visible, but most curves remain within a relatively close range. Around 2030, most national electricity demand forecasts fall within approximately 4,300 to 5,100 TWh, while a small number of high-growth pathways exceed 5,300 TWh. Divergence during this period mainly reflects differences in institutional assumptions about near-term economic growth, electrification speed and emerging load growth, but the projections remain broadly within the same order of magnitude. In contrast, as the forecast horizon extends toward 2050, the gap among curves expands substantially. By 2050, the forecasts are mostly located around 5,700 to 7,200 TWh, with some upper-bound pathways approaching 10,000 TWh. This indicates that uncertainty in U.S. national electricity demand forecasts

has a clear cumulative time effect: the longer the forecast horizon, the more strongly differences in scenario assumptions are amplified.

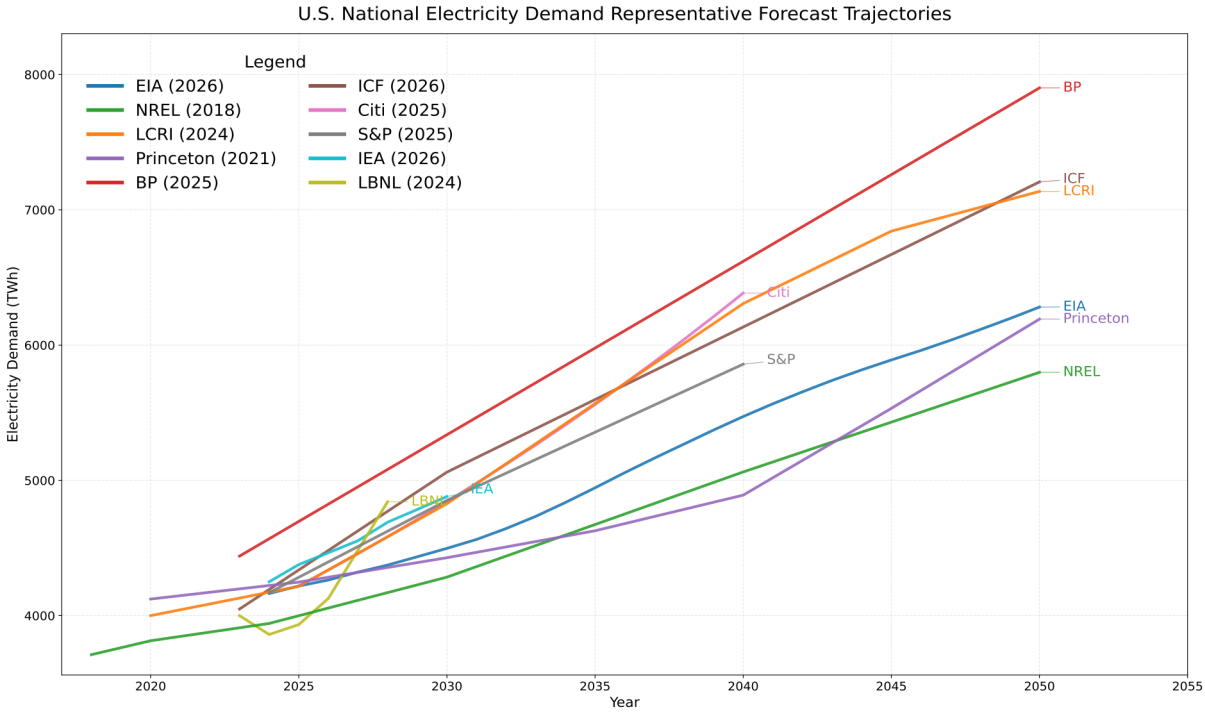


Figure 5.1.2 National Electricity Demand Representative Forecast Trajectories

Among all the forecasts, EIA provides an important official benchmark in Figure 5.1.1. Its counterfactual baseline with sensitivity cases forms a relatively systematic range of policy and market sensitivities, representing a moderate growth pathway. The NREL curves place greater emphasis on combinations of electrification levels and technology advancement assumptions. NREL's Reference scenarios are generally more conservative, while High Electrification scenarios are clearly higher. Princeton and LCRI are more closely associated with long-term energy transition and net-zero pathway analysis, and their high-value scenarios reflect the upward pressure on long-term electricity demand from deep electrification and low-carbon energy system expansion. Some institutions show a stronger tendency toward high-growth projections. BP, ICF and LCRI are higher than most baseline forecasts in the long term, especially after 2040. These projections assume stronger end-use electrification, higher energy-transition-related loads, or faster electricity demand growth.

Figure 5.1.2, which keeps one representative trajectory per institution, shows that different sources do not share the same forecast horizon. EIA, NREL, Princeton, LCRI, ICF, and BP extend to 2050 and are therefore useful for observing demand divergence under long-term energy system transition, while IEA, LBNL, Citi, and S&P Global are concentrated before 2040, making them more suitable for interpreting near-term and medium-term demand changes. Since conditions and markets are changing rapidly, long-term forecasts should be understood as scenario-based references rather than precise numerical predictions. In comparison, forecasts around 2030 and 2035 have more direct relevance for current planning decisions.

5.1.2 Data Center Electricity Demand Forecast Trajectories

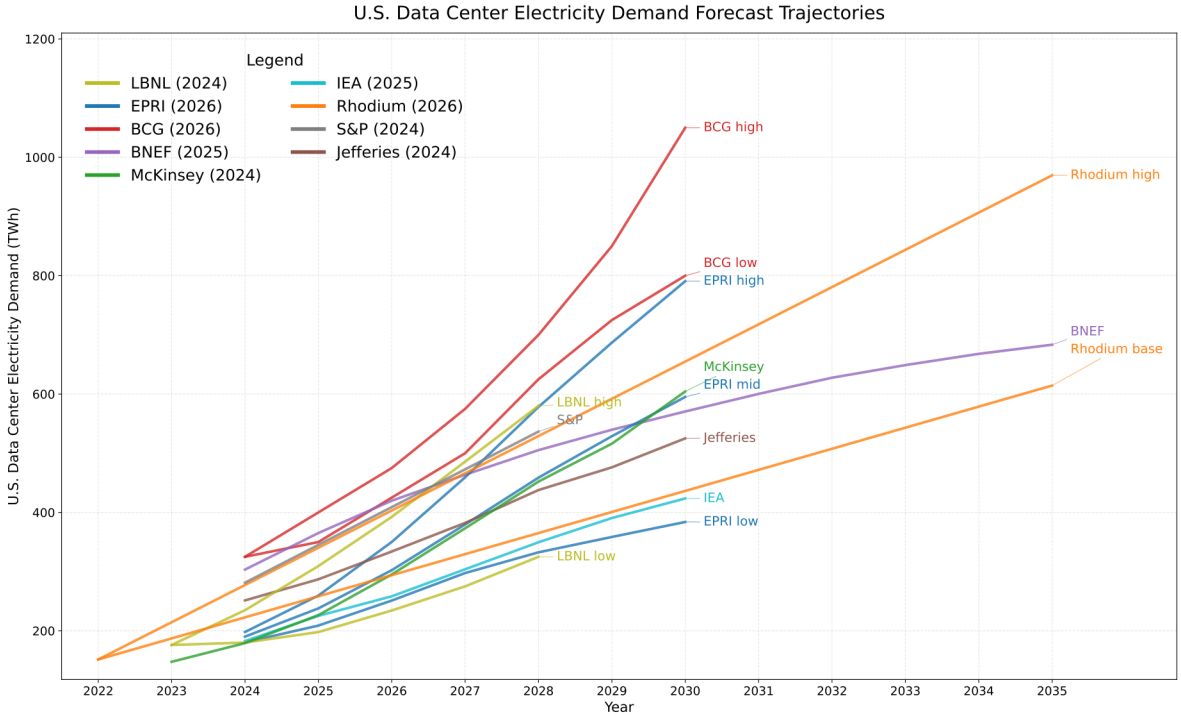


Figure 5.1.3 Data Center Electricity Demand Forecast Full Trajectories

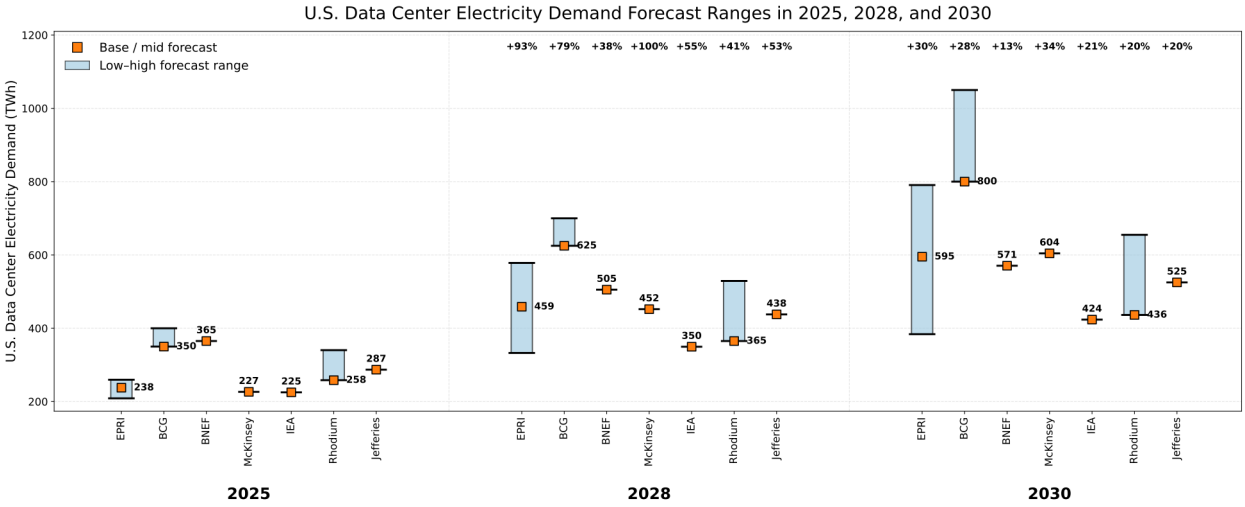


Figure 5.1.4 Data Center Electricity Demand Forecast Ranges in 2025, 2028 and 2030

Figure 5.1.3 presents multi-institutional forecast trajectories for U.S. data center electricity demand. Compared with national electricity demand forecasts, data center forecasts have a much shorter time horizon, but they show faster growth and earlier divergence. Most sources focus their projections around 2028 or 2030, suggesting that publicly comparable data center electricity demand forecasts remain concentrated in the near-term to medium-term period. It essentially shows that, driven by the expansion of AI, U.S. data center electricity demand is shifting from a relatively limited specialized load category into an emerging high-growth load that can significantly affect power system planning. From a temporal perspective, the historical demand between 2022 and 2025 recorded by different institutions already shows some variation. Moving toward 2030, the divergence among curves

expands rapidly. By 2028, LBNL's low and high scenarios indicate that U.S. data center electricity demand may reach approximately 325 to 580 TWh. By 2030, most forecasts are concentrated around 400 to 650 TWh, while some high-growth scenarios may reach approximately 1,000 TWh. Figure 5.1.4 compares the institutional forecast ranges in 2025, 2028 and 2030.

From an institutional perspective, LBNL provides an important data center-specific benchmark. Its low-end and high-end scenarios clearly show the potential lower and upper bounds of U.S. data center electricity demand in the coming years. EPRI's low, medium, and high scenarios also reflect uncertainty in data center growth from a power-sector research perspective. BCG and Rhodium show stronger high-growth tendencies, suggesting that under assumptions of rapid AI infrastructure expansion, data center electricity demand may increase substantially within a short period. In contrast, sources such as IEA, Jefferies, BloombergNEF, McKinsey, and S&P Global mainly provide single projected or medium scenarios, representing estimates of data center electricity demand under specific target-year or market-based assumptions.

A comparison between national electricity demand forecasts and data center electricity demand forecasts reveals two different uncertainty structures. National demand shows long-term divergence from a much larger baseline, mainly driven by assumptions about electrification, macroeconomic growth, and low-carbon transition pathways. Data center demand shows rapid growth and near-term divergence from a smaller baseline, mainly driven by the pace of AI infrastructure deployment, expansion of computing demand and local grid interconnection constraints.

Although data center electricity demand remains much smaller than total U.S. electricity demand in absolute terms, its rapid growth suggests that it may account for a larger share in the future and become increasingly important. In addition, data center loads are often large-scale, highly concentrated and continuously operating, making them important for utility investment, grid expansion, transmission interconnection and generation resource planning.

5.2 Forecast Range Analysis Results

Forecast range analysis summarizes each institution's forecast results for a specific year as a low–high interval and marks the central value, preparing for a more direct comparison of forecast levels and uncertainty ranges across institutions within the same target year.

5.2.1 National Electricity Demand Forecast Ranges

a. National Electricity Demand Forecast Ranges in 2030

Figure 5.2.1 presents the forecast ranges for U.S. national electricity demand in 2030. NREL provides the lowest central estimate at 4,267 TWh; Princeton and EIA are slightly higher at 4,428 TWh and 4,562 TWh; LCRI, Citi, S&P Global, and IEA are clustered around 4,800 to 4,900 TWh; and ICF and BP are positioned at the higher end, with central values beyond 5,000 TWh. This distribution suggests that most institutional forecasts still fall within a relatively comparable range by 2030, but a clear three-tier ordering of demand expectations has already emerged.

The uncertainty range within each institution also differs. LCRI shows the widest low–high range in 2030, indicating a broad set of possible demand outcomes under its low-carbon transition assumptions. NREL and Princeton also present visible forecast ranges, reflecting the influence of different electrification pathways and transition assumptions. Conversely, Citi, IEA, ICF, and BP appear mainly as single or near-single forecast values, making them more suitable as point estimates rather than full uncertainty ranges.

Overall, national electricity demand forecasts by 2030 have a common range around 4,800 TWh that most forecasts agree on, while the differences between forecasts remain moderate.

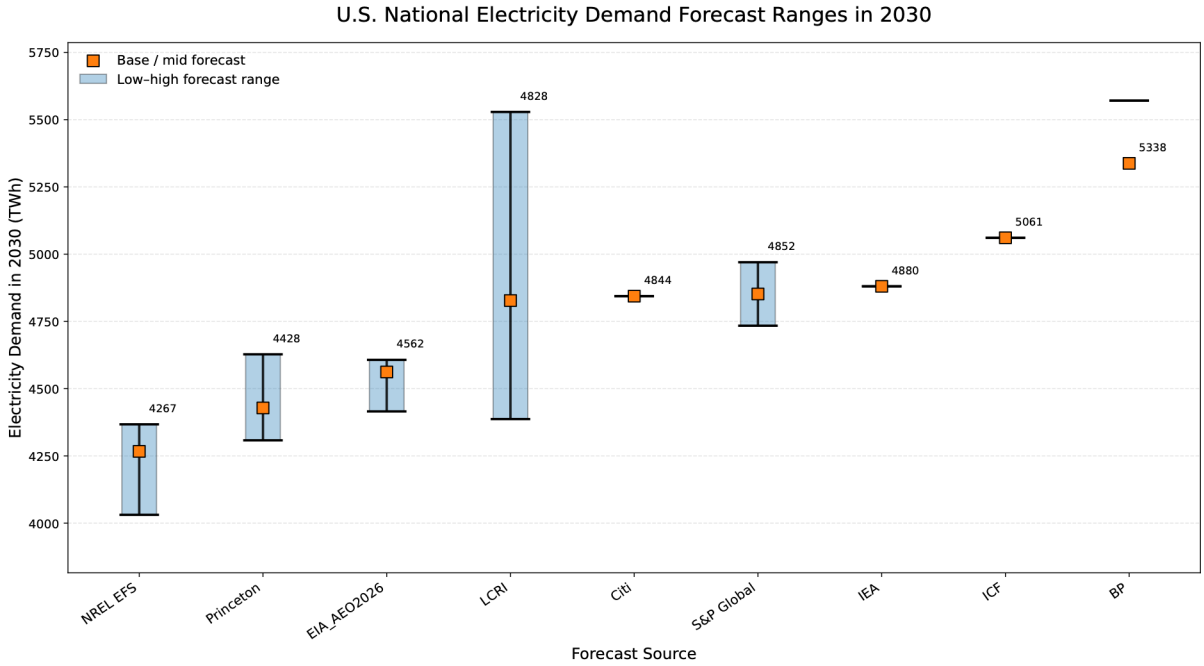


Figure 5.2.1 National Electricity Demand Forecast Ranges in 2030

b. National Electricity Demand Forecast Ranges in 2035

Figure 5.2.2 presents the forecast ranges for U.S. national electricity demand in 2035. By this year, the institutional interpretation of demand growth becomes more differentiated. Princeton and NREL EFS remain at the lower end, at approximately 4,600 TWh; EIA rises to approximately 4,937 TWh; S&P Global, Citi, LCRI and ICF are concentrated around 5,300 to 5,600 TWh; and BP is still the highest, with a central forecast of 5,979 TWh.

The internal forecast ranges also become more informative in 2035. LCRI shows the widest range, extending from a relatively moderate demand pathway to a much higher upper bound. This indicates that LCRI's long-term transition assumptions create a broad uncertainty interval even by the mid-2030s. NREL and Princeton also display relatively wide ranges, but their central values remain lower, suggesting more conservative representative pathways with higher-demand possibilities under alternative scenarios. EIA shows a narrower range, indicating a more moderate set of sensitivity outcomes around its central outlook.

Overall, national electricity demand forecasts by 2035 have a common range around 5,500 TWh that most forecasts agree on, while the differences between forecasts increase.

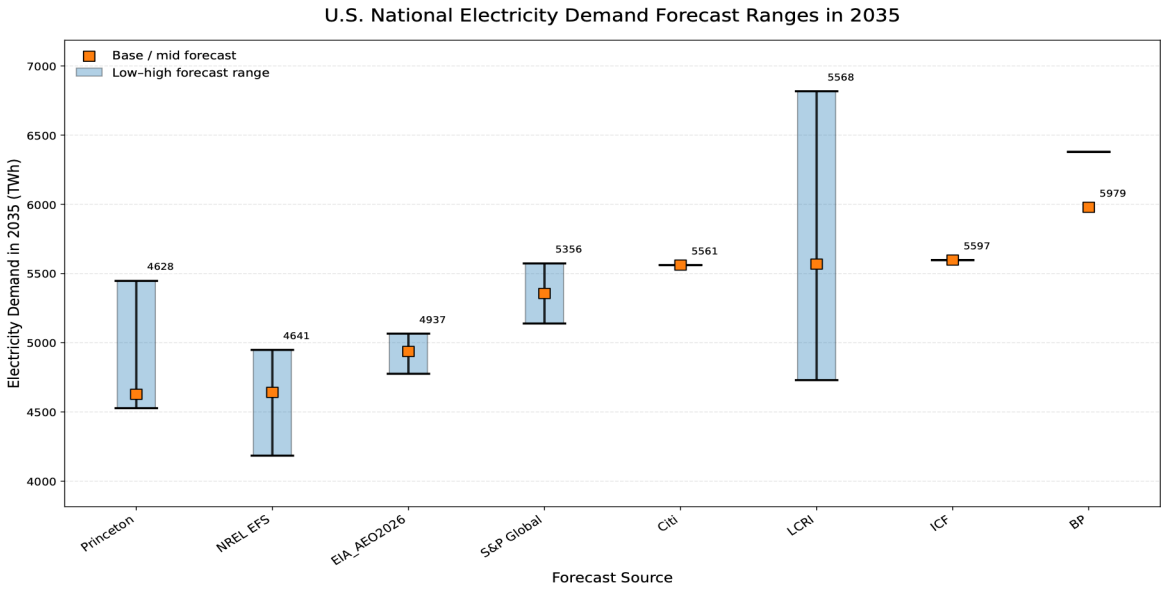


Figure 5.2.2 National Electricity Demand Forecast Ranges in 2035

c. National Electricity Demand Forecast Ranges in 2050

Figure 5.2.3 presents the forecast ranges for U.S. national electricity demand in 2050. This year shows the widest divergence among the selected benchmark years. NREL's forecast is located at 5,727 TWh; EIA is 5,995 TWh; Princeton is 6,192 TWh; LCRI and ICF are higher, with central forecasts at around 7,200 TWh; and BP provides the highest central forecast of all institutions, at 7,902 TWh.

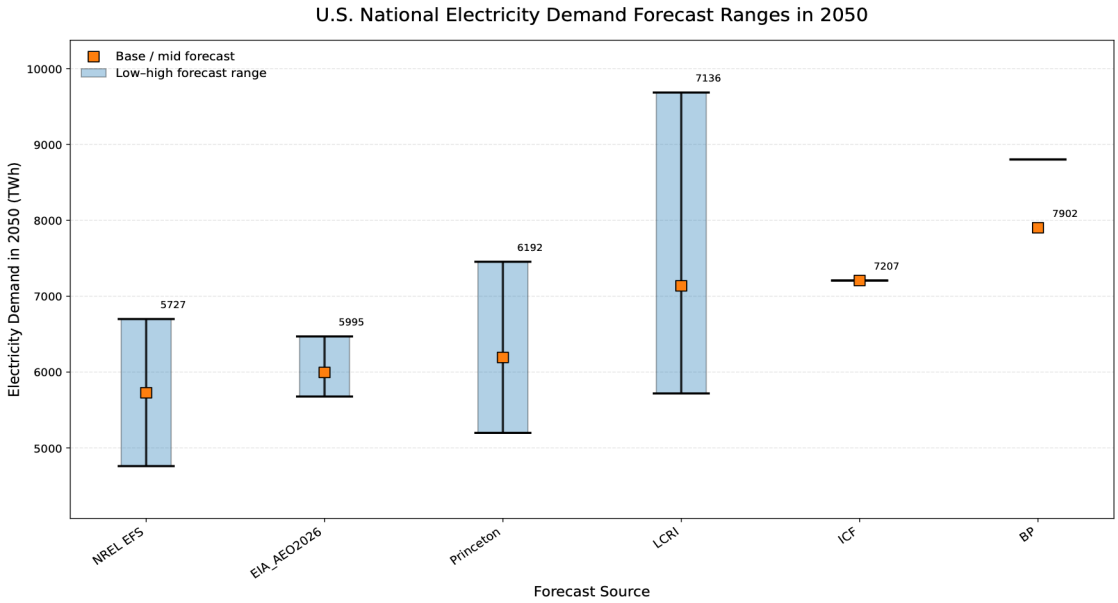


Figure 5.2.3 National Electricity Demand Forecast Ranges in 2050

The most important feature of the 2050 range plot is the expansion of uncertainty intervals. NREL, Princeton, and especially LCRI show broad low–high ranges, indicating that long-term electricity

demand is highly uncertain. LCRI's upper bound is particularly high, approaching 10,000 TWh, which is far above most central forecasts, suggesting that under aggressive low-carbon transition or high-electrification assumptions, U.S. national electricity demand could be higher than conventional baseline projections. Overall, national electricity demand forecasts by 2050 have a common range around 5,700 to 7,200 TWh that most forecasts agree on, while the differences between forecasts reach their highest.

5.2.2 Data Center Electricity Demand Forecast Ranges

a. Data Center Electricity Demand Forecast Ranges in 2028

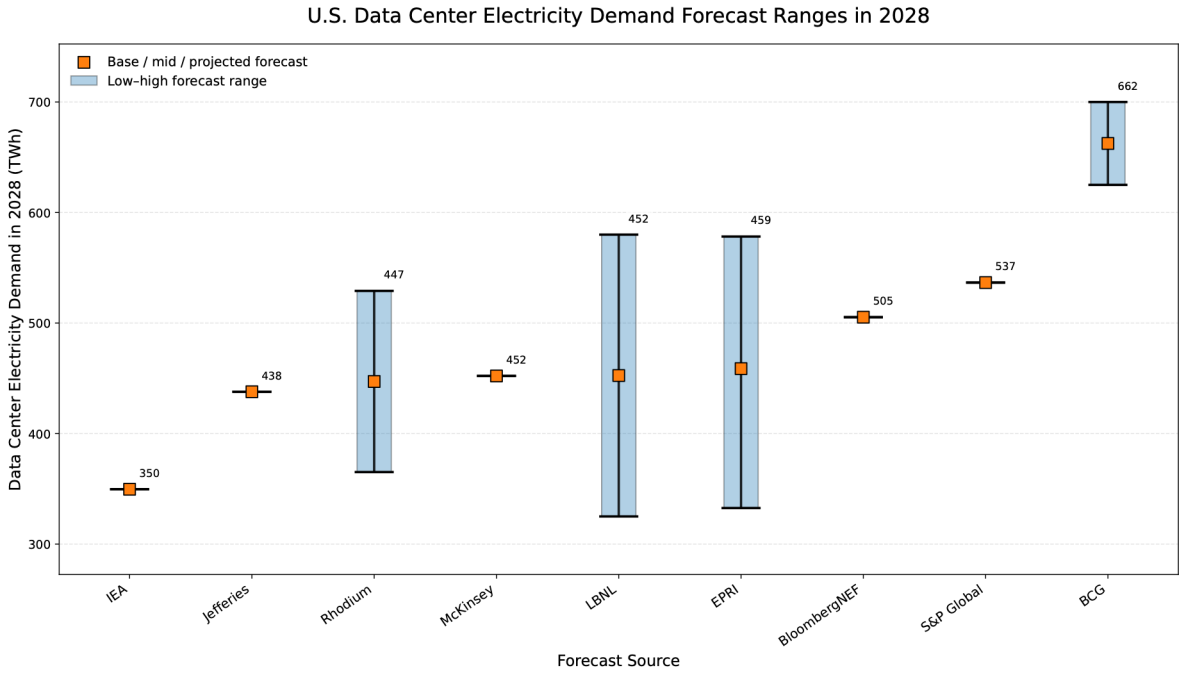


Figure 5.2.4 Data Center Electricity Demand Forecast Ranges in 2028

Figure 5.2.4 presents the forecast ranges for U.S. data center electricity demand in 2028. IEA is positioned at the lowest end at 350 TWh; Jefferies is at 438 TWh; Rhodium, McKinsey, LBNL, and EPRI are clustered in the middle range, with representative forecasts around 450 TWh; BloombergNEF and S&P Global are higher, at 505 TWh and 537 TWh, respectively; and BCG provides the highest central estimate at 662 TWh. This indicates that by 2028, U.S. data center electricity demand forecasts already reach about 1.7 to 3 times their 2022–2024 starting values, reflecting a strong sense of growth.

From the perspective of internal institutional ranges, Rhodium, LBNL, EPRI, and BCG all provide visible low–high forecast ranges, while IEA, Jefferies, McKinsey, BloombergNEF, and S&P Global mainly appear as single or near-single projected values. The ranges provided by LBNL and EPRI are especially important because they represent a data center-specific research perspective and a power-sector research perspective, respectively. Their ranges suggest that even within technical or industry-oriented analytical frameworks, 2028 data center electricity demand remains highly dependent on assumptions about AI adoption, server deployment scale, utilization rates, cooling demand and grid interconnection capacity. BCG has a more aggressive view of near-term data center load expansion and gives the highest forecast.

Overall, data center electricity demand forecasts by 2028 have a common range around 450 TWh that most forecasts agree on, with the uncertainty already significant before 2030 and the divergence of forecasts beginning to increase.

b. Data Center Electricity Demand Forecast Ranges in 2030

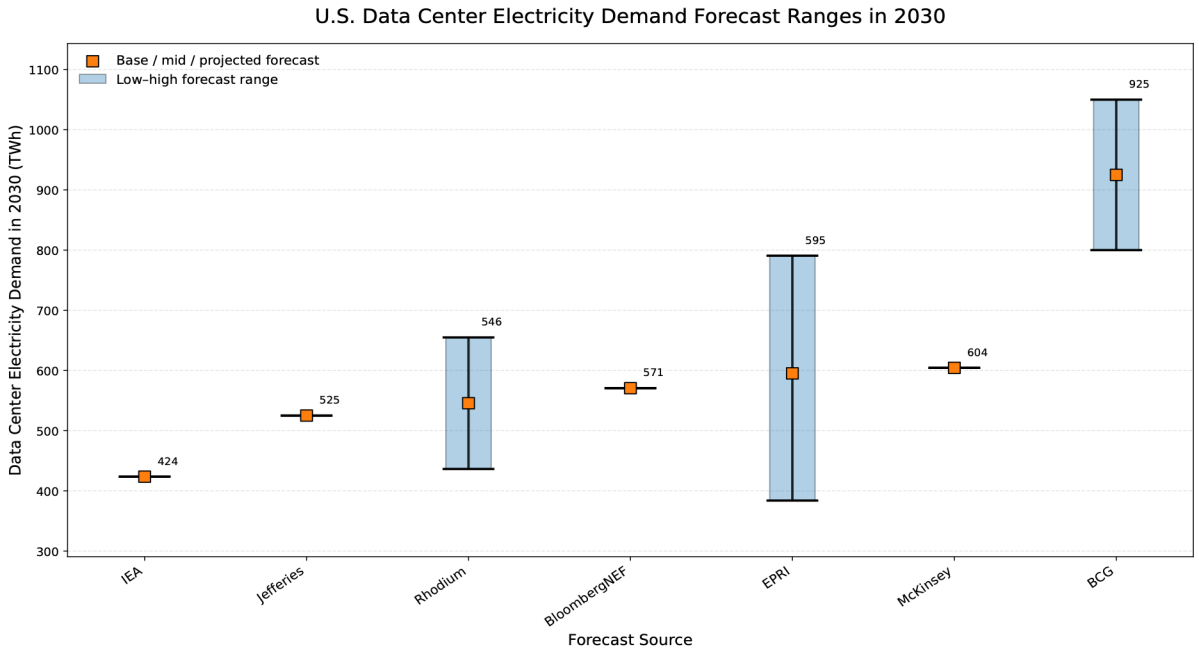


Figure 5.2.5 Data Center Electricity Demand Forecast Ranges in 2030

Figure 5.2.5 presents the forecast ranges for U.S. data center electricity demand in 2030. Compared with 2028, the divergence across institutions becomes even more pronounced. IEA remains at the lower end at 424 TWh; Jefferies, Rhodium, BloombergNEF, EPRI and McKinsey form a middle forecast group, with central values around 520 to 600 TWh; and BCG is clearly positioned at the highest end, at approximately 925 TWh. It can be seen that by 2030, institutional forecasts range from roughly 400 TWh to nearly 1,000 TWh, which is roughly twice to even four times their 2022–2024 starting values. From the perspective of forecast ranges, Rhodium, EPRI, and BCG provide clear low–high intervals in 2030, while IEA, Jefferies, BloombergNEF and McKinsey mainly appear as point estimates. EPRI's range is particularly wide, extending from below 400 TWh to nearly 800 TWh, suggesting a broad range of possible outcomes from the perspective of power system planning. Rhodium also shows meaningful internal uncertainty, while BCG's range remains high even at its lower bound and extends beyond 1,000 TWh at the upper bound. It is clear that the forecast divergence in 2030 exists not only across institutions, but also within institutions that explicitly define multiple scenarios.

Overall, data center electricity demand forecasts by 2030 have a common range around 600 TWh that most forecasts agree on, with the divergence of forecasts reaching its highest.

5.3 Difference Heatmap Analysis Results

Difference heatmaps directly quantify the differences between institutions or scenarios, allowing the analysis to identify which forecasts are relatively close, which institutions deviate most strongly from others, and how forecast divergence changes across target years.

5.3.1 National Central Forecast Differences

a. National Central Forecast Differences in 2030

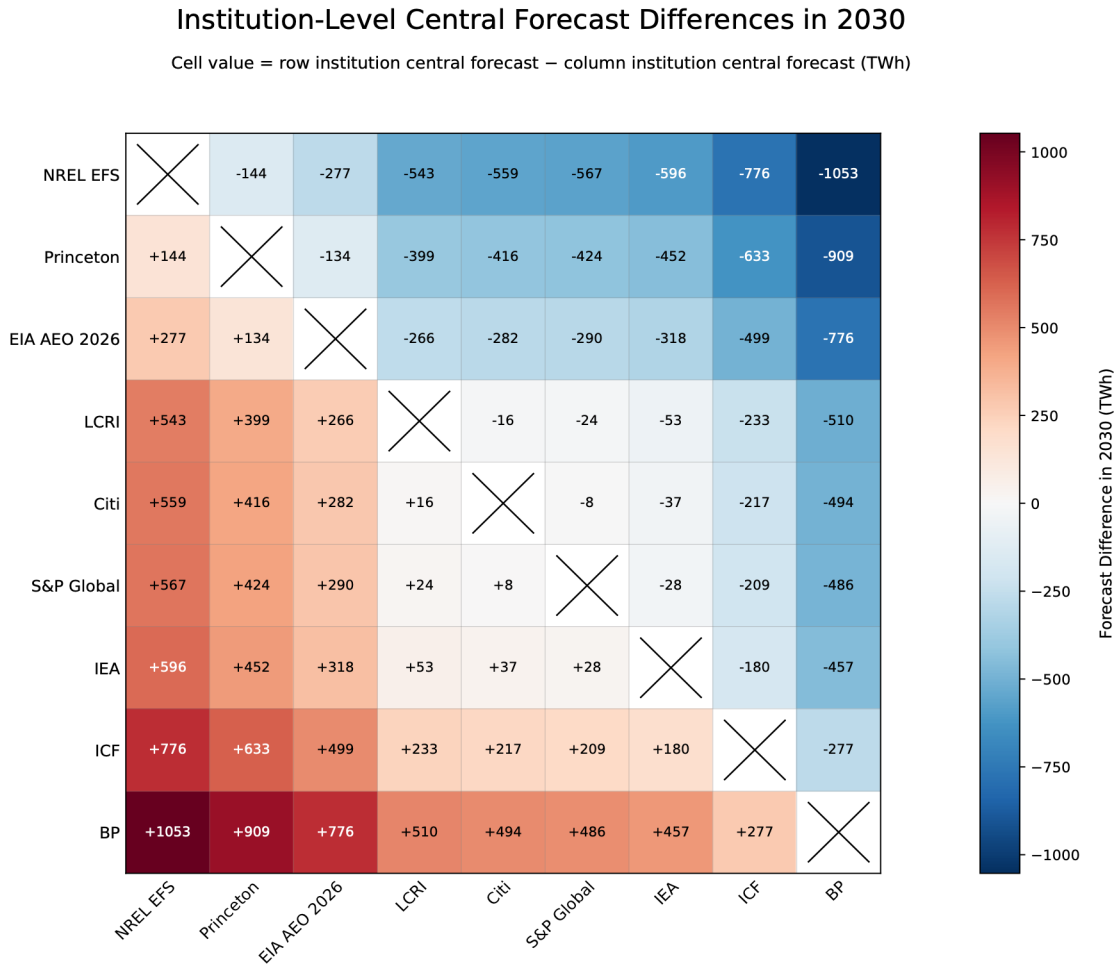


Figure 5.3.1 National Central Forecast Differences in 2030

Figure 5.3.1 presents inter-institutional differences among central forecasts for U.S. national electricity demand in 2030. NREL EFS, Princeton and EIA are located in the lower forecast group; LCRI, Citi, S&P Global and IEA form the middle forecast group; and ICF and BP are in the higher forecast group. The largest inter-institutional difference in 2030 is between NREL and BP at 1,053 TWh, compared to the smallest difference between S&P Global and Citi at only 8 TWh.

The middle forecast group is especially close internally, indicating that although institutions already differ in their assessment of 2030 electricity demand growth, several sources show strong agreement within the middle forecast range. In contrast, a large difference suggests that near-term to medium-term forecasts are already being shaped by different growth assumptions.

Most difference values are within an acceptable range from 300 TWh to 400 TWh, about 6%–8% of total electricity demand in 2030. Therefore, the 2030 results indicate that national electricity demand forecasts have begun to separate into different levels, but have not yet reached the more extreme divergence observed in longer-term projections.

b. National Central Forecast Differences in 2035

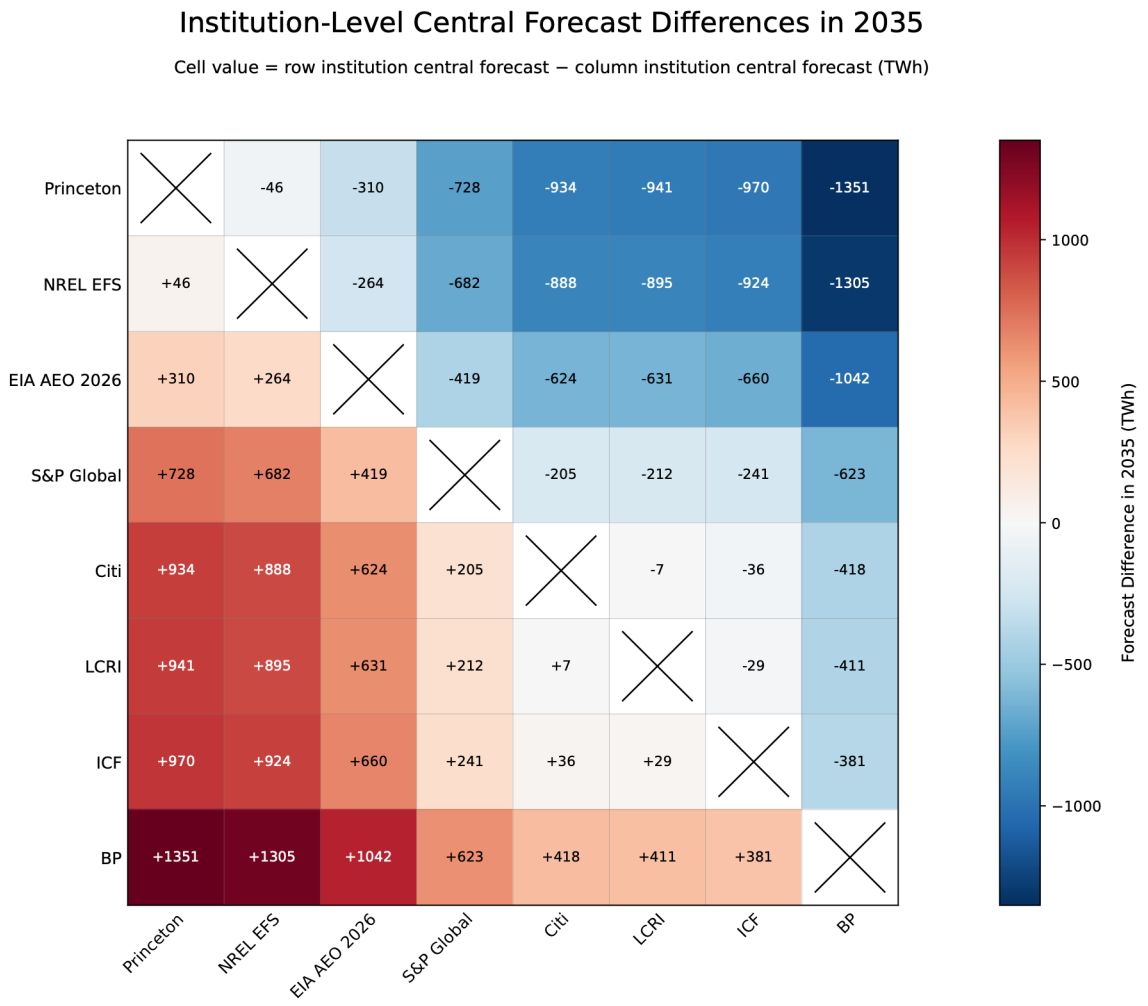


Figure 5.3.2 National Central Forecast Differences in 2035

Figure 5.3.2 presents inter-institutional differences among central forecasts for U.S. national electricity demand in 2035. Compared with 2030, the differences are noticeably larger. Princeton, NREL and EIA remain at the lower end; S&P Global is in the middle; Citi, LCRI and ICF form a closely clustered middle-high forecast group; and BP is still at the highest end. The difference between the lowest source, Princeton, and the highest source, BP, reaches 1,351 TWh, with NREL and EIA differing from BP by 1,305 TWh and 1,042 TWh, respectively.

The middle-high forecast group is highly concentrated, but its distance from the lower forecast group has expanded substantially. Citi and LCRI differ by only about 7 TWh, LCRI and ICF differ by about 29 TWh, and Citi and ICF differ by about 36 TWh. This suggests that by 2035, institutional forecasts are not randomly dispersed; instead, they form clearer forecast clusters. Some institutions project higher medium-term to long-term electricity demand growth, while others maintain more conservative central forecasts.

The number of forecast difference values falling into the acceptable range of approximately 400 TWh decreases, but the boundaries between different scenarios and institutional outlooks are clearer in 2035 and it is easier to classify those forecasts to identify how aggressive they are.

c. National Central Forecast Differences in 2050

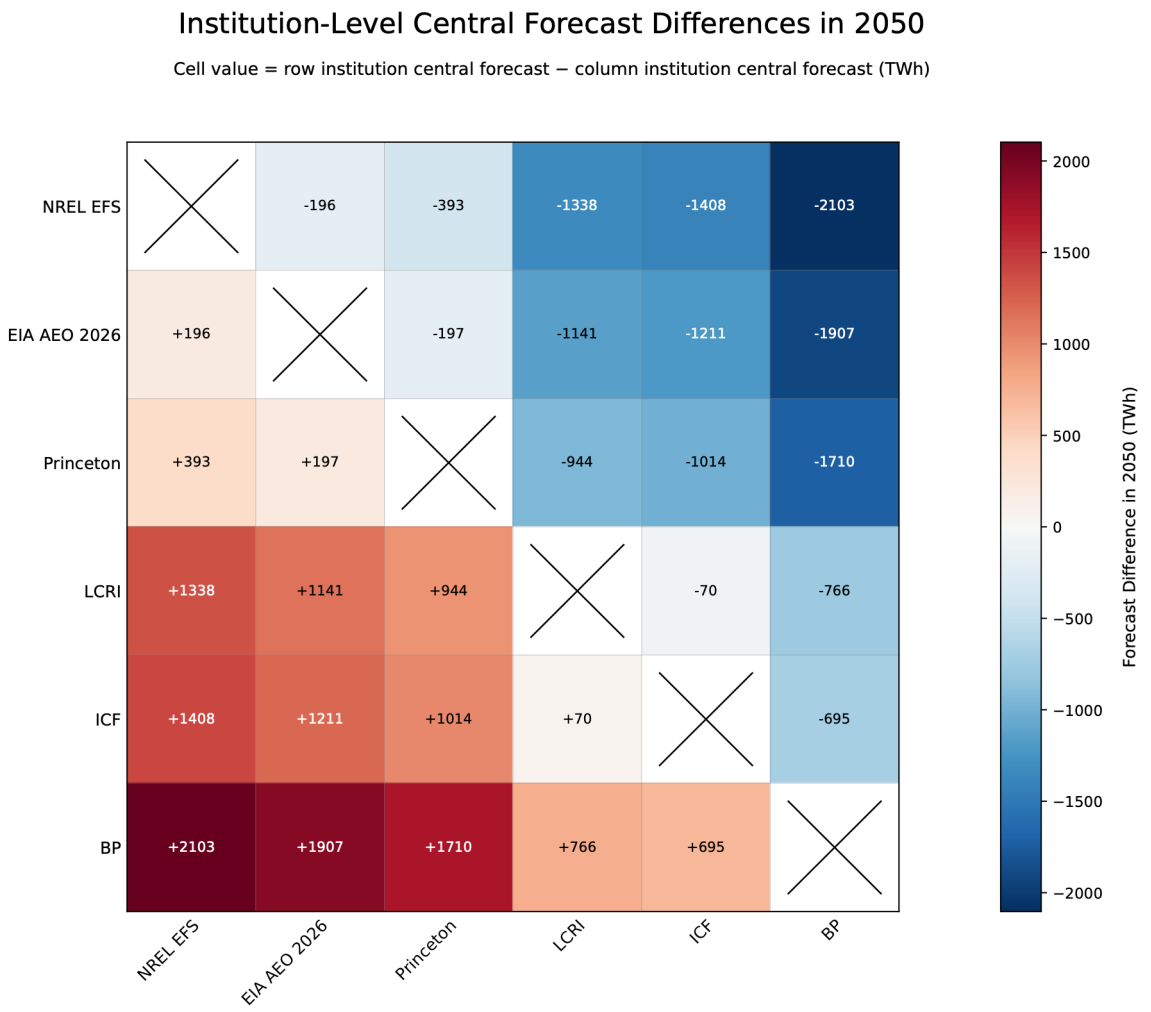


Figure 5.3.3 National Central Forecast Differences in 2050

Figure 5.3.3 presents inter-institutional differences among central forecasts for U.S. national electricity demand in 2050. The divergence is most pronounced in 2050, with inter-institutional differences reaching the scale of thousands of TWh and the largest difference exceeding 2,000 TWh. NREL, EIA and Princeton form the lower-to-middle forecast group; LCRI and ICF form the high forecast group; and BP is positioned at the highest end. BP exceeds NREL by 2,103 TWh, EIA by about 1,907 TWh, and Princeton by about 1,710 TWh.

Within-group differences also reveal institutional similarities. NREL, EIA and Princeton are relatively close to one another: NREL differs from EIA by 196 TWh, while EIA differs from Princeton by 197 TWh. On the higher end, LCRI and ICF differ by only 70 TWh, indicating that their 2050 central forecasts are highly similar. The most significant differences occur between the lower-to-middle forecast group and the high forecast group, with BP showing the largest deviation from other institutions.

These results show that long-term national electricity demand forecasts have evolved from the layered structure observed in the near-term and medium-term period into a more pronounced structural divergence. Differences across institutions in 2050 reflect different visions of the future U.S. energy system. The lower group corresponds to more moderate long-term demand growth, while the high forecast group implies a more transformative increase in electricity demand.

5.3.2 Data Center Central Forecast Differences

a. Data Center Central Forecast Differences in 2028

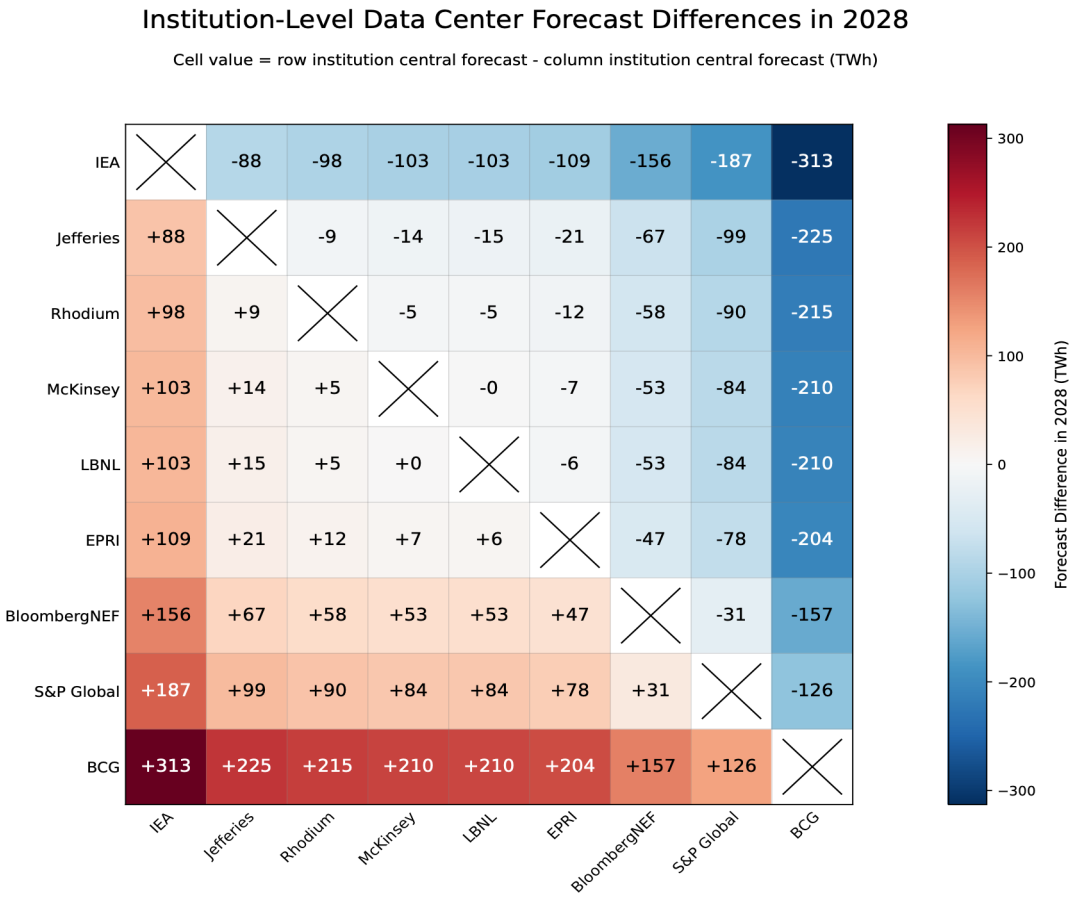


Figure 5.3.4 Data Center Central Forecast Differences in 2028

Figure 5.3.4 presents inter-institutional differences among central forecasts for U.S. data center electricity demand in 2028. Overall, the 2028 results already show clear stratification across institutions, although most differences remain within the range of tens to several hundred TWh. IEA is positioned at the lowest end, while BCG is positioned at the highest end. The difference between these two is 313 TWh, representing the largest difference in 2028. Jefferies, Rhodium, McKinsey, LBNL and EPRI form a relatively tight middle forecast group, while BloombergNEF and S&P Global are positioned in the middle-high range.

Within the middle forecast group, the differences are very small. Rhodium differs from McKinsey by only 5 TWh, Rhodium differs from LBNL also by 5 TWh, McKinsey and LBNL are almost identical, and LBNL differs from EPRI by only 6 TWh. This indicates that several institutions have reached relatively

similar assessments of data center electricity demand in 2028, clustering around 450 TWh. In contrast, BCG is clearly higher than all other sources: it exceeds S&P Global by 126 TWh, BloombergNEF by 157 TWh, and the middle forecast group by more than 200 TWh.

A small number of difference values are within an acceptable range of 50 TWh, about 11% of data center electricity demand in 2028. These results show that data center central forecasts had already begun to diverge by then. Most institutions are concentrated within a middle forecast band, suggesting some degree of consensus regarding near-term data center load growth.

b. Data Center Central Forecast Differences in 2030

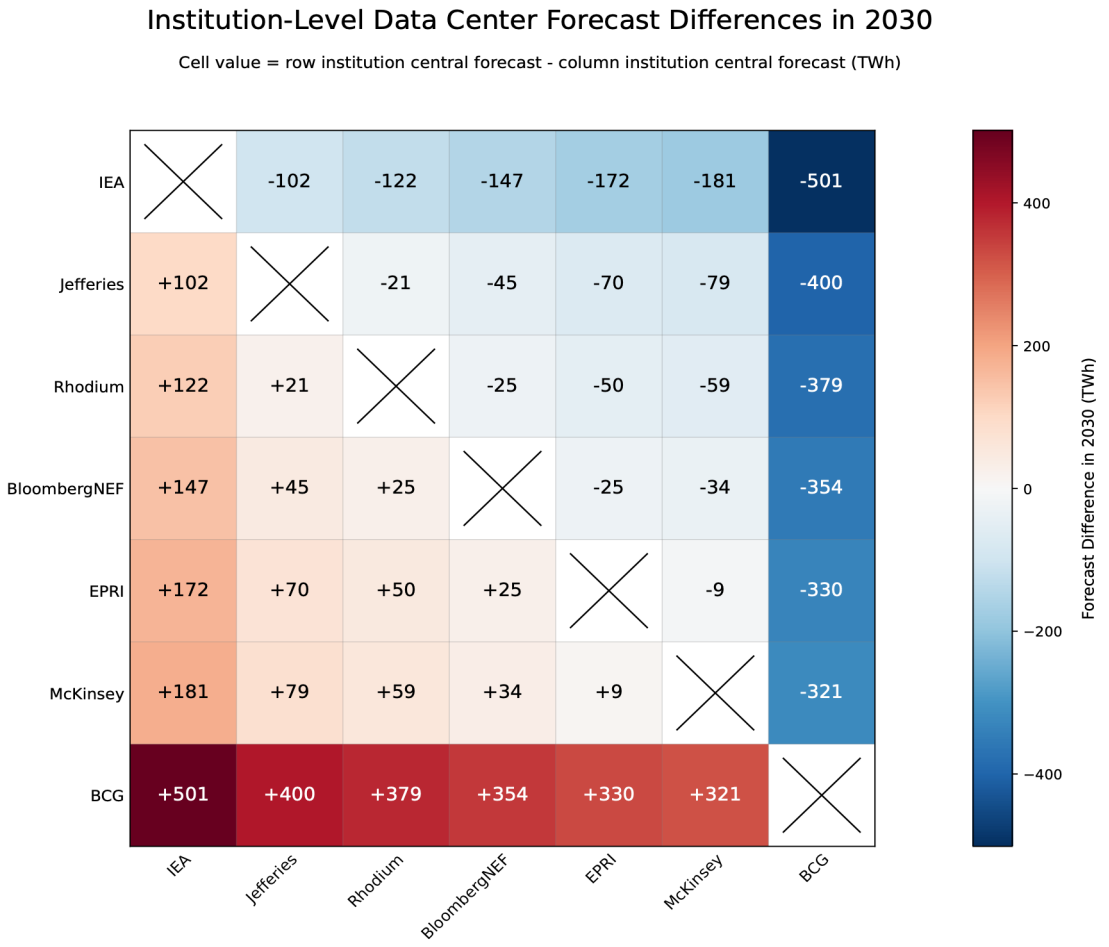


Figure 5.3.5 Data Center Central Forecast Differences in 2030

Figure 5.3.5 presents inter-institutional differences among central forecasts for U.S. data center electricity demand in 2030. Compared with 2028, the differences across institutions expand further. IEA remains at the lowest end, while BCG remains at the highest end. The difference between the two reaches 501 TWh, higher than the maximum difference in 2028. Jefferies, Rhodium, BloombergNEF, EPRI and McKinsey form the middle forecast group, although this group also begins to show a clearer internal gradient.

The middle forecast group remains relatively close internally, but its differences are larger than in 2028. Jefferies differs from Rhodium by 21 TWh, Rhodium differs from BloombergNEF by 25 TWh, BloombergNEF differs from EPRI by 25 TWh, and EPRI differs from McKinsey by 9 TWh. This indicates

that these institutions still occupy a relatively consistent middle forecast band in 2030, but their forecasts now form a more continuous ordering from lower to higher values. BCG shows the most pronounced deviation from other institutions: its central forecast is approximately 321 TWh higher than McKinsey, 330 TWh higher than EPRI, 354 TWh higher than BloombergNEF, and 400 TWh higher than Jefferies.

Only a few difference values are within an acceptable range of 60 TWh, which is 10% of data center electricity demand in 2030. These results indicate that inter-institutional divergence in data center electricity demand forecasts becomes more pronounced by 2030. The middle forecast group still shows some degree of agreement, but high-growth projections have separated more clearly from the rest of the field.

5.4 Within-Institution Scenario Differences

Within-institution scenario differences focus on the uncertainty generated by each institution's own scenario design. Since not all institutions provide multiple scenarios, and some only provide a simple low–high range, this section analyzes the most representative institutional cases.

5.4.1 National Within-Institution Scenario Differences

a. LCRI

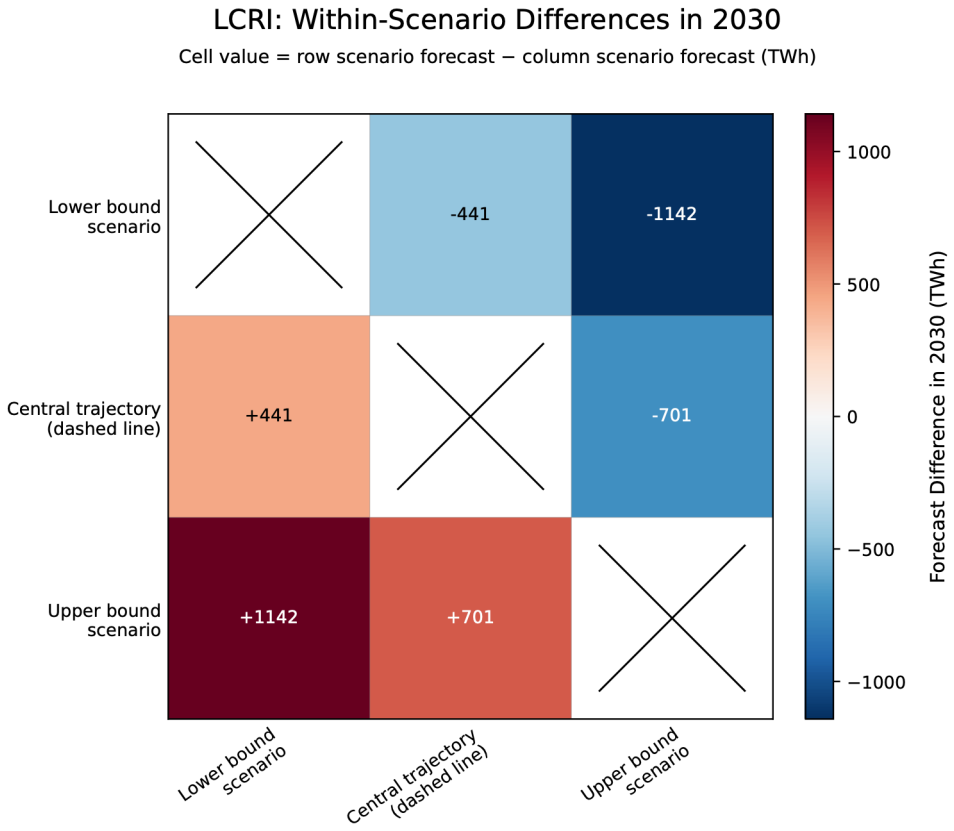


Figure 5.4.1 LCRI Forecast Scenario Heatmap

Figure 5.4.1 presents the forecast differences among LCRI scenarios in 2030. LCRI shows very large internal scenario differences. The lower-bound scenario differs from the upper-bound scenario by 1,142 TWh; the central trajectory differs from the upper-bound scenario by 701 TWh; and the lower-bound scenario differs from the central trajectory by 441 TWh. Within the LCRI framework, different low-carbon transition pathways and electricity demand growth assumptions alone are sufficient to generate differences of more than 1,000 TWh in the 2030 national electricity demand forecast.

This result shows that LCRI's uncertainty is not only reflected in its difference from other institutions, but also in the width of its own scenario structure. For national electricity demand forecasting, LCRI represents broad transition-pathway uncertainty: different assumptions about the intensity of energy transition, end-use electrification speed, and low-carbon resource deployment can substantially change the projected scale of future electricity demand.

b. EIA

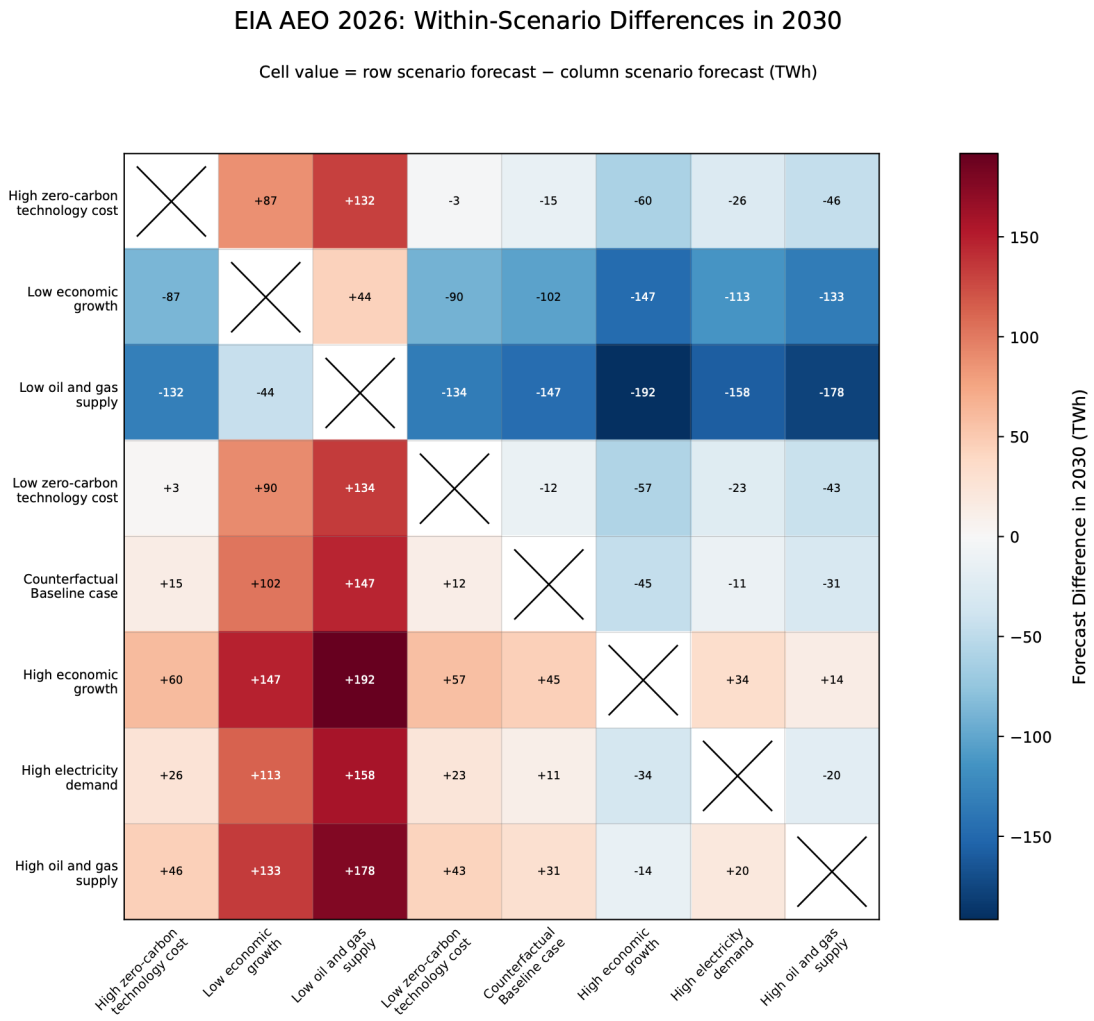


Figure 5.4.2 EIA Forecast Scenario Heatmap

Figure 5.4.2 presents the forecast differences among EIA AEO 2026 sensitivity cases in 2030. Compared with LCRI, EIA shows much smaller internal scenario differences. The largest difference

among its sensitivity cases is 192 TWh, and most scenario differences are concentrated within the range of several tens to more than one hundred TWh.

This indicates that the EIA sensitivity cases function more as bounded deviations around the official baseline rather than as fundamentally different energy system transition pathways. The internal differences mainly reflect the effects of macroeconomic growth, oil and gas supply, zero-carbon technology costs, and high electricity demand assumptions on near-term to medium-term electricity demand. Compared with LCRI, EIA's scenario range is narrower and is more suitable as a sensitivity reference around an official benchmark forecast.

c. NREL

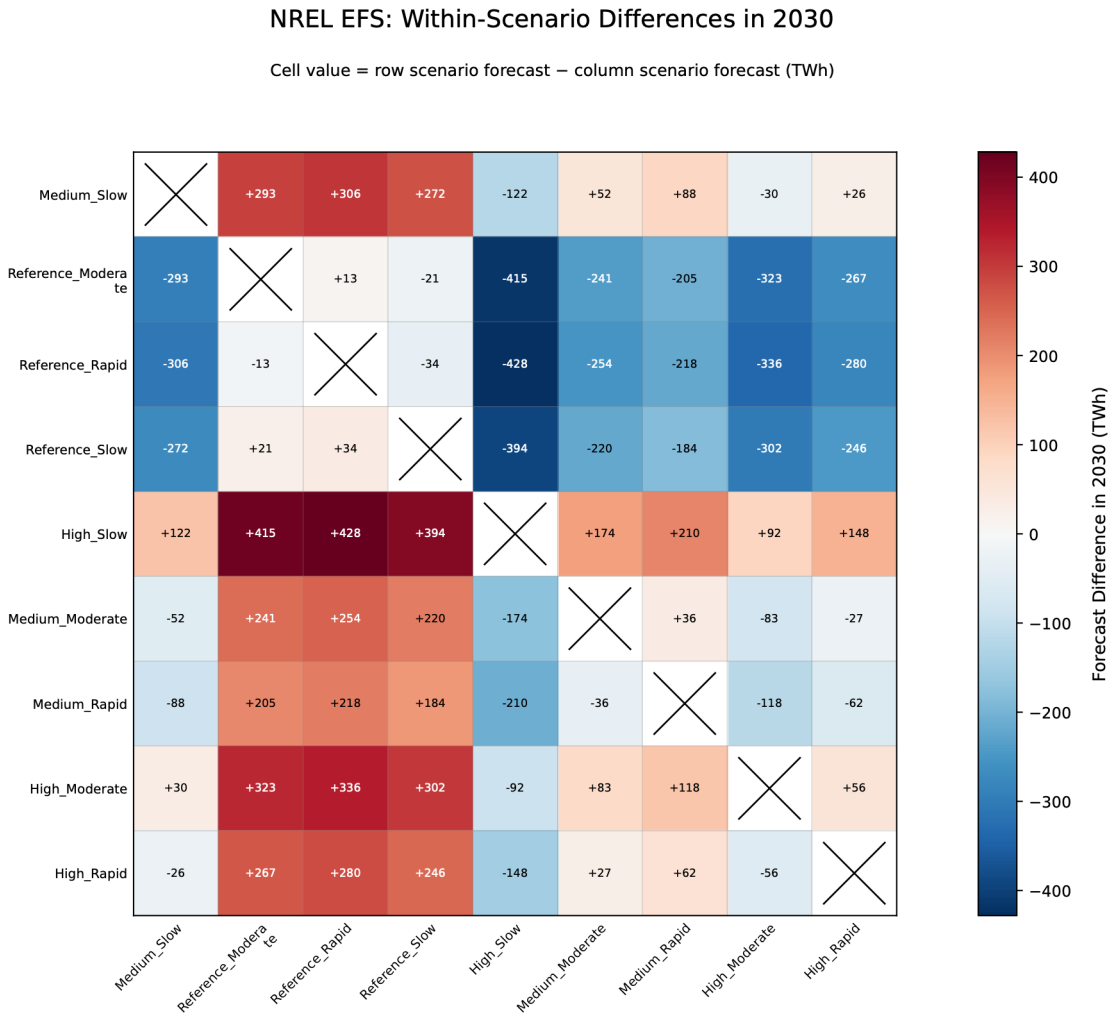


Figure 5.4.3 NREL Forecast Scenario Heatmap

Figure 5.4.3 presents the forecast differences of all NREL scenarios in 2030. NREL has a more complex internal scenario structure since it combines different electrification levels with different technology advancement assumptions. In 2030, two NREL scenario differences exceed 400 TWh, showing that within the NREL EFS framework, the interaction between electrification level and technology advancement assumptions can affect national electricity demand projections. Compared with EIA, NREL has larger internal scenario differences, reflecting the importance of electrification pathways in

shaping demand outcomes. Compared with LCRI, however, its internal spread in 2030 remains more limited. NREL therefore represents a moderate-to-large level of internal uncertainty generated by a multi-scenario modeling structure.

d. Other National Sources

Other national sources have simpler internal scenario structures, but they still reflect a certain degree of forecast uncertainty. Princeton's low, mid, and high scenarios differ by up to approximately 320 TWh in 2030. S&P Global's lower and upper outlooks differ by approximately 236 TWh, while BP's Below 2° and Current Trajectory scenarios differ by approximately 233 TWh.

These results indicate that some institutions do not provide scenario matrices as complex as EIA or NREL, nor do they show a transition-pathway range as wide as LCRI, but they still express internal forecast uncertainty through low–high or two-scenario structures.

5.4.2 Data Center Within-Institution Scenario Differences

a. EPRI

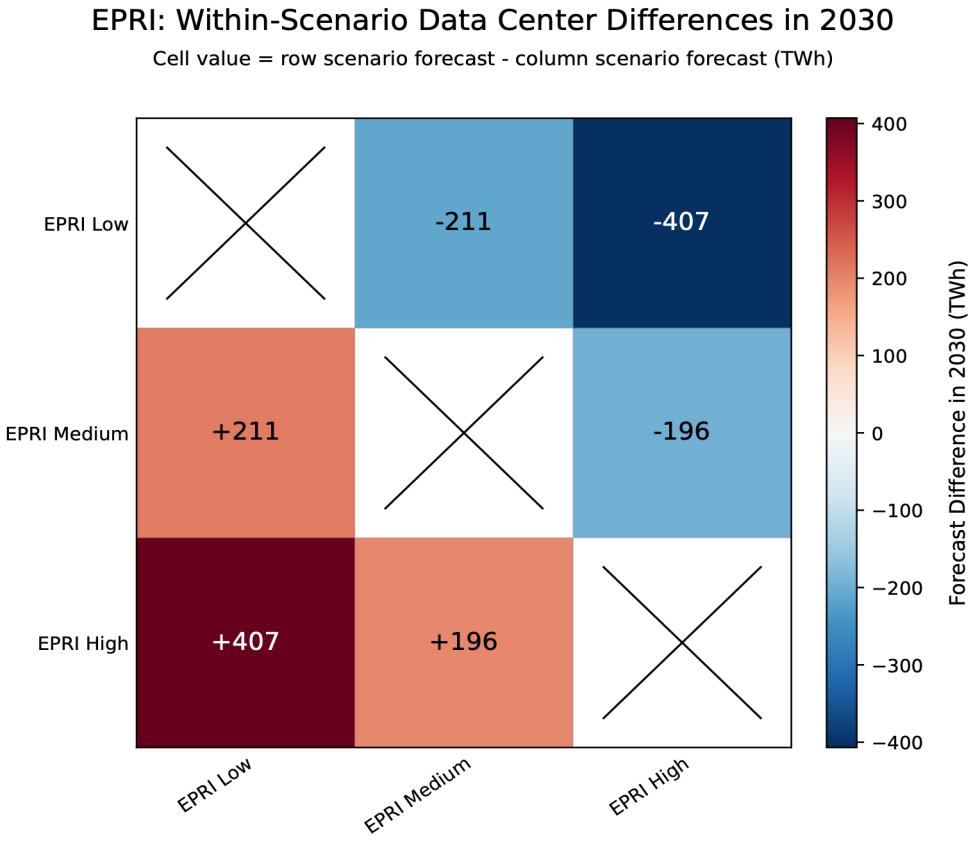


Figure 5.4.4 EPRI Data Center Scenario Heatmap

Figure 5.4.4 presents the forecast differences among EPRI data center electricity demand scenarios in 2030. EPRI's three scenarios show the following differences: Low differs from High by 407 TWh; Low differs from Medium by 211 TWh; and Medium differs from High by 196 TWh.

This result indicates that EPRI provides a very wide possible range for 2030 data center electricity demand. From a power system planning perspective, an internal scenario difference of about 400 TWh is large enough to affect utility investment, grid expansion, generation resource planning and

transmission interconnection decisions. EPRI's result therefore shows that data center load forecasts can contain strong uncertainty even within a single power-sector research framework.

b. BCG

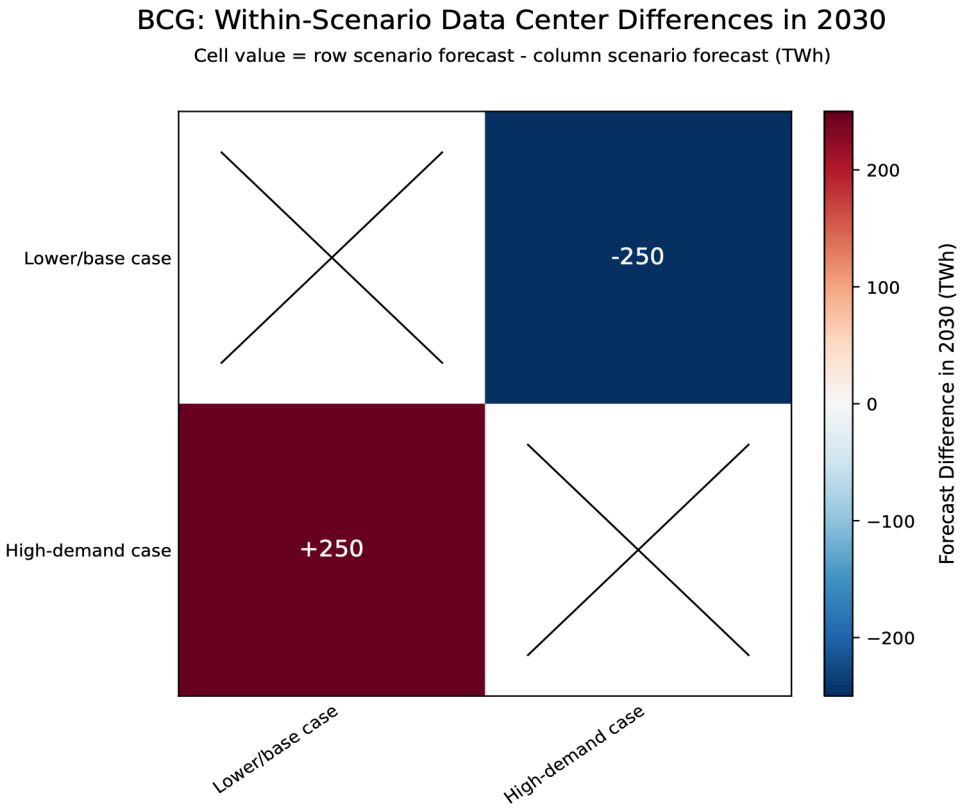


Figure 5.4.5 BCG Data Center Scenario Heatmap

Figure 5.4.5 presents the forecast differences between BCG's data center electricity demand scenarios in 2030. BCG's high-demand case is approximately 250 TWh higher than its base case, which means that, within its analytical framework, adopting a more aggressive AI infrastructure expansion assumption can change the 2030 data center electricity demand forecast. Since BCG is already positioned as a high-forecast source in the inter-institutional comparison, its high-demand case further reinforces a high-growth outlook for rapidly increasing data center load. BCG's internal difference therefore represents additional upward uncertainty within an already high-growth forecast perspective.

c. Rhodium

Figure 5.4.6 presents the forecast difference between Rhodium's baseline data center growth and high data center growth scenarios in 2030. Rhodium's high data center growth scenario is 219 TWh higher than its baseline data center growth scenario. This difference is smaller than EPRI's low–high range, but it is still significant. It shows that changing only the data center growth assumption can generate a forecast difference of more than 200 TWh in 2030. Rhodium's result indicates that uncertainty in data center electricity demand forecasts is highly concentrated in assumptions about

the strength of AI-driven load growth, rather than only in long-term energy system transition assumptions.

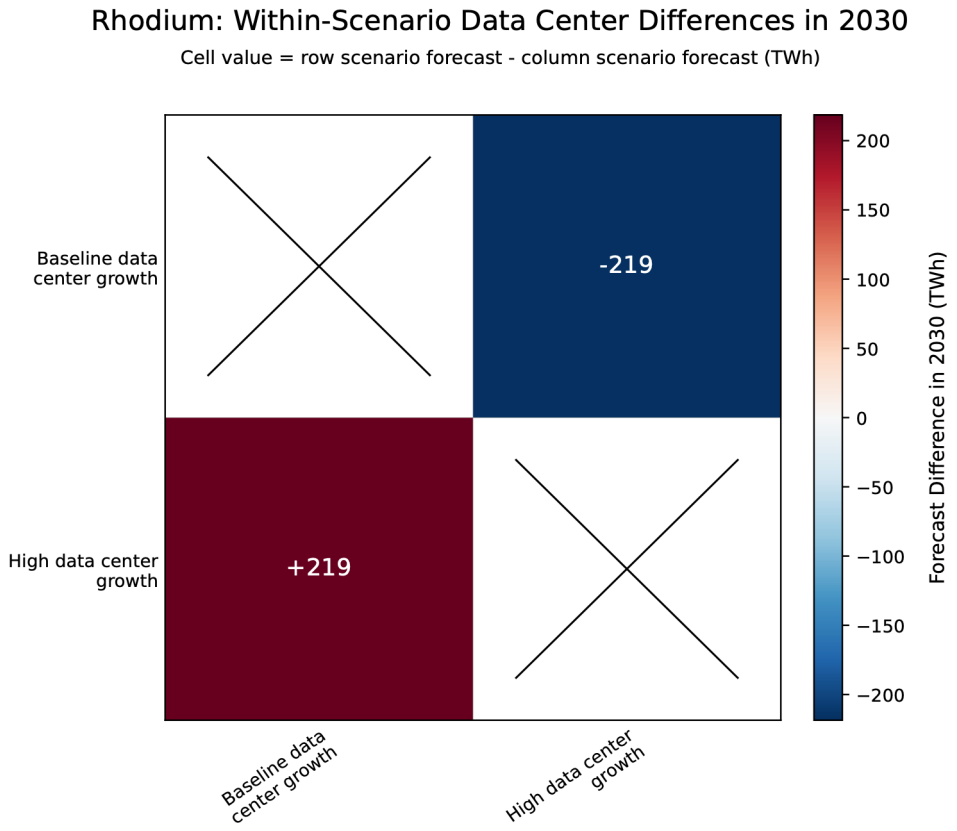


Figure 5.4.6 Rhodium Data Center Scenario Heatmap

5.5 Forecast Level and Growth Intensity Ranking Results

This study also uses representative forecast value, growth multiple, and compound annual growth rate (CAGR) as the main comparative indicators to convert institutional forecasts into ranking metrics, to clarify which institutions project higher demand in selected target years, and which institutions imply faster demand growth over time.

5.5.1 National Target-Year Forecast Level Ranking

Table 5.1 summarizes the national forecast level ranking for target years 2030, 2035 and 2050. This ranking is based on each institution's representative forecast value in the target year. A higher rank indicates a higher projected national electricity demand level in that year.

Table 5.1 National Electricity Demand Forecast Level Ranking (2030, 2035, 2050)

Ranking	2030 Source Institution	2030 Value (TWh)	2035 Source Institution	2035 Value (TWh)	2050 Source Institution	2050 Value (TWh)
1	BP	5,454.22	BP	6,178.67	BP	8,352.00
2	ICF	5,060.90	ICF	5,597.35	ICF	7,206.70
3	IEA	4,880.50	LCRI	5,568.14	LCRI	7,136.48

Ranking	2030 Source Institution	2030 Value (TWh)	2035 Source Institution	2035 Value (TWh)	2050 Source Institution	2050 Value (TWh)
4	S&P Global	4,852.00	Citi	5,561.05	Princeton	6,192.41
5	Citi	4,843.86	S&P Global	5,356.00	EIA	5,995.21
6	LCRI	4,827.55	EIA	4,937.11	NREL	5,827.20
7	EIA	4,562.01	NREL	4,685.12	—	—
8	Princeton	4,428.29	Princeton	4,627.51	—	—
9	NREL	4,290.29	—	—	—	—

The 2030 ranking shows that BP is positioned at the highest end at 5,454 TWh, followed by ICF at 5,061 TWh. IEA, S&P Global, Citi, and LCRI are clustered around 4,800 TWh, forming a relatively dense middle forecast range. EIA and Princeton are positioned at the lower end, while NREL provides the lowest representative value, at only 4,290 TWh.

The 2035 ranking shows that BP remains at the highest end at 6,179 TWh. ICF, LCRI and Citi follow BP and are very close to one another, at around 5,560 to 5,600 TWh. S&P Global occupies an intermediate position at 5,356 TWh, while EIA, NREL and Princeton remain at the lower end. Compared with 2030, the distance between high-forecast and low-forecast sources increases further in 2035, suggesting that medium-term national electricity demand forecasts begin to show more pronounced institutional divergence.

The 2050 ranking further reinforces the long-term divergence structure. BP remains at the highest end at 8,352 TWh, while ICF and LCRI follow at around 7,200 TWh, forming the long-term high-forecast group. Princeton, EIA and NREL are positioned at the lower end, with values around 6,000 TWh. The difference between the highest and lowest long-term representative values exceeds 2,500 TWh, indicating that forecast differences across institutions are significantly amplified over longer time horizons.

Generally, BP, ICF and LCRI give more aggressive forecasts, while NREL, EIA and Princeton are more conservative.

5.5.2 National Growth Multiple and CAGR Ranking

Table 5.2 summarizes the national growth intensity ranking for target years 2030, 2035 and 2050. This ranking is based on CAGR, while the table also reports each source's growth multiple. Since different institutions have different available starting years, CAGR should be interpreted as the implied annual growth intensity within each source's own forecast window, rather than as a completely uniform comparison from a common base year.

Table 5.2 National Electricity Demand Growth Multiple and CAGR Ranking (2030, 2035, 2050)

Ranking	2030 Source	2030 Growth Multiple	2030 CAGR (%)	2035 Source	2035 Growth Multiple	2035 CAGR (%)	2050 Source	2050 Growth Multiple	2050 CAGR (%)
1	ICF	1.250	3.24	Citi	1.318	2.80	BP	1.881	2.37
2	BP	1.228	2.98	BP	1.392	2.79	ICF	1.780	2.16
3	Citi	1.148	2.80	ICF	1.383	2.74	LCRI	1.784	1.95

Ranking	2030 Source	2030 Growth Multiple	2030 CAGR (%)	2035 Source	2035 Growth Multiple	2035 CAGR (%)	2050 Source	2050 Growth Multiple	2050 CAGR (%)
4	S&P Global	1.164	2.56	S&P Global	1.284	2.30	NREL	1.570	1.42
5	IEA	1.148	2.33	LCRI	1.392	2.23	Princeton	1.502	1.37
6	LCRI	1.207	1.90	EIA	1.149	1.40	EIA	1.395	1.34
7	NREL	1.156	1.22	NREL	1.263	1.38	—	—	—
8	EIA	1.061	1.20	Princeton	1.122	0.77	—	—	—
9	Princeton	1.074	0.72	—	—	—	—	—	—

The CAGR ranking does not fully match the target-year forecast level ranking, because CAGR measures growth intensity rather than the absolute forecast level in the target year. In the 2030 ranking, ICF has the highest implied CAGR at approximately 3.24%; BP ranks second at approximately 2.98%; and Citi ranks third at approximately 2.80%. S&P Global and IEA also show relatively high near-term to medium-term growth intensity, at 2.56% and 2.33%, respectively. In contrast, LCRI, NREL, EIA and Princeton have lower CAGRs.

In the 2035 ranking, Citi ranks first, with a CAGR of approximately 2.80%, which is consistent with its forecast construction based on a 2.8% CAGR assumption. BP and ICF follow closely, with CAGRs of approximately 2.79% and 2.74%, respectively. S&P Global and LCRI rank next, with CAGRs of approximately 2.30% and 2.23%. EIA, NREL and Princeton have the lowest CAGRs, with Princeton still below 1%. In terms of medium-term growth intensity, Citi, BP and ICF represent stronger sustained growth assumptions, while Princeton, NREL and EIA are closer to moderate growth pathways.

The 2050 ranking shows that BP has the highest long-term CAGR at 2.37%; ICF ranks second at approximately 2.16%; and LCRI ranks third, at approximately 1.95%. NREL, Princeton and EIA have the lowest long-term CAGRs around 1.40%. This result indicates that BP, ICF and LCRI not only have higher absolute forecast levels in 2050, but also imply stronger long-term average annual growth intensity, while EIA, Princeton and NREL represent more moderate long-term growth trajectories.

5.5.3 Data Center Target-Year Forecast Level Ranking

Table 5.3 summarizes the representative forecast level ranking for U.S. data center electricity demand in target years 2028 and 2030. Similar to the national electricity demand ranking, the representative value for each institution is determined using the same curve-selection framework: if only one curve is available, that curve is used directly; if two curves are available, the representative value is calculated as the average of the two curves; and if three or more curves are available, the central or base trajectory is selected whenever possible.

Table 5.3 U.S. Data Center Electricity Demand Forecast Level Ranking (2028, 2030)

Ranking	2028 Source Institution	2028 Value (TWh)	2030 Source Institution	2030 Value (TWh)
1	BCG	662.50	BCG	925.00
2	S&P Global	536.62	McKinsey	604.38
3	BloombergNEF	505.29	EPRI	595.17
4	EPRI	458.75	BloombergNEF	570.59
5	LBNL	452.50	Rhodium	545.64

Ranking	2028 Source Institution	2028 Value (TWh)	2030 Source Institution	2030 Value (TWh)
6	McKinsey	452.12	Jefferies	525.13
7	Rhodium	447.12	IEA	423.62
8	Jefferies	437.83	—	—
9	IEA	349.61	—	—

The 2028 ranking shows that the highest forecast levels are already substantially separated from more moderate ones. Institutions such as BCG, S&P Global and BloombergNEF occupy the highest positions, indicating that some expect U.S. data center electricity demand to exceed 500 TWh by 2028. By contrast, more moderate projections from sources such as IEA, McKinsey, Rhodium and Jefferies remain at comparatively lower levels. It indicates that forecast divergence in data center electricity demand emerges much earlier than in national demand, even within a relatively short forecast horizon.

The 2030 ranking further reinforces this divergence structure. Only BCG projects U.S. data center electricity demand to exceed 800 TWh, while all other projections remain below this level. Compared with the 2028 ranking, the spread between higher and lower forecasts expands further in 2030, suggesting that uncertainty surrounding AI-driven data center expansion increases rapidly over time. Another important observation is that institutions with similar 2028 values may diverge significantly by 2030. This suggests that the key difference among data center forecasts is not only the estimated current scale of electricity consumption, but also the assumed pace of future growth. Forecasts associated with aggressive AI infrastructure deployment, higher rack densities, faster hyperscale expansion, and stronger computing demand tend to move toward the upper end of the ranking, while forecasts assuming slower deployment, stronger efficiency improvements, or tighter infrastructure constraints remain relatively lower.

5.5.4 Data Center Growth Multiple and CAGR Ranking

Table 5.4 summarizes the U.S. data center electricity demand growth intensity ranking for 2028 and 2030, similar to the national electricity demand analysis.

Table 5.4 U.S. Data Center Electricity Demand Growth Multiple and CAGR Ranking (2028, 2030)

Ranking	2028 Source Institution	2028 Growth Multiple	2028 CAGR (%)	2030 Source Institution	2030 Growth Multiple	2030 CAGR (%)
1	McKinsey	3.063	25.09	McKinsey	4.094	22.31
2	EPRI	2.413	24.63	EPRI	3.130	20.95
3	LBNL	2.571	20.79	BCG	2.846	19.04
4	Rhodium	2.951	19.76	Rhodium	3.601	17.37
5	BCG	2.038	19.49	IEA	2.319	15.05
6	IEA	1.914	17.62	Jefferies	2.089	13.07
7	S&P Global	1.908	17.52	BloombergNEF	1.880	11.10
8	Jefferies	1.742	14.89	—	—	—
9	BloombergNEF	1.665	13.60	—	—	—

The 2028 CAGR ranking shows that U.S. data center electricity demand forecasts imply exceptionally rapid growth rates compared with national electricity demand forecasts. Several institutions project annualized growth rates above typical national electricity demand growth assumptions, reflecting expectations of accelerated AI infrastructure deployment and rapidly increasing computational demand.

The 2030 CAGR ranking further demonstrates the structural difference between data center electricity demand growth and broader national electricity demand growth. While national electricity demand forecasts in this study generally imply long-term CAGR values of approximately 1% to 3%, many data center electricity demand forecasts imply substantially higher annual growth intensity over much shorter time horizons. This indicates that data center electricity demand growth is occurring at a structurally different scale and speed from broader national electricity demand growth.

The growth ranking also shows that target-year forecast level and CAGR are related but not identical indicators. Some institutions rank highly because they project extremely large target-year electricity demand values, while others rank highly because they imply faster relative growth from lower starting points. Therefore, growth intensity ranking provides additional insight beyond absolute forecast magnitude alone. In particular, some forecasts with moderate target-year electricity demand levels still imply very aggressive annual growth rates, reflecting assumptions of rapid short-term expansion.

5.5.5 Integrated Ranking Analysis

The ranking analysis reveals a clear structural difference between U.S. national electricity demand growth and U.S. data center electricity demand growth.

In the national electricity demand rankings, institutional differences expand gradually over longer forecast horizons, with divergence becoming most pronounced by 2050. Although institutions differ in their assumptions regarding electrification, economic growth, low-carbon transition pathways, and technology deployment, most national electricity demand forecasts remain within a relatively bounded long-term range.

In the data center electricity demand rankings, the divergence emerges much earlier and is stronger. Significant differences across institutional forecasts are already visible before 2030, both in target-year electricity demand levels and in implied annual growth intensity. Compared with national electricity demand forecasts, data center electricity demand forecasts imply higher CAGR values over much shorter forecast horizons, indicating that AI-driven data center expansion is occurring at a structurally different growth scale and speed.

The ranking results also demonstrate that forecast magnitude and growth intensity are related but distinct dimensions of forecast uncertainty. Some institutions project very large target-year electricity demand values because they assume rapid AI infrastructure deployment and hyperscale expansion, while others imply extremely high growth rates even from relatively smaller starting values. Therefore, comparing only target-year electricity demand levels is insufficient for understanding forecast divergence. Growth intensity analysis provides an additional perspective on how institutions interpret the pace of future electricity demand expansion.

Overall, the ranking analysis confirms that AI-driven data center load growth is becoming one of the most dynamic and uncertain components of future U.S. electricity demand growth. While national electricity demand forecasts primarily reflect long-term structural transition pathways, data center electricity demand forecasts are rapidly evolving. Consequently, future U.S. electricity demand should

be interpreted not as a single deterministic pathway, but as a structured range of possibilities shaped by institutional assumptions, technological uncertainty and evolving AI infrastructure development.

6. Discussion and Conclusion

This chapter synthesizes the analytical results of the study and discusses their implications for AI-driven U.S. electricity demand forecasting. The central finding is that national electricity demand and data center electricity demand are both projected to increase in the coming years, but their forecasts exhibit different uncertainty structures. National forecasts mainly reflect long-term structural transition uncertainty, while data center forecasts reflect short-horizon, technology-driven and infrastructure-constrained uncertainty. This distinction is important because AI-driven data center load growth introduces a faster, more concentrated and more volatile source of demand growth than conventional loads.

6.1 Interpretation of Forecast Differences

The forecast differences observed in this study arise primarily from differences in institutional objectives, modeling boundaries, scenario assumptions and time horizons. Institutions did not produce forecasts under a unified modeling framework, while aiming for different research purposes and reflecting different assumptions regarding economic growth, electrification pathways, energy-system transition, AI deployment, technology development, infrastructure expansion and policy direction. Therefore, the differences represent different interpretations of how the future U.S. electricity system may evolve under changing conditions.

For U.S. national electricity demand forecasts, the observed divergence mainly reflects different long-term interpretations of structural energy-system transition. More conservative forecasts generally imply slower electrification, lower economic growth, more limited expansion of emerging electricity loads, or stronger efficiency improvements. By contrast, higher-growth forecasts often imply more aggressive electrification pathways, larger-scale low-carbon transition, stronger industrial electricity demand growth, expanded hydrogen production, and faster growth in electricity-intensive sectors. The gradual expansion of divergence over time indicates that uncertainty in national demand is primarily cumulative and long-horizon in nature, reflecting increasingly different assumptions about the long-term evolution of the U.S. energy system.

U.S. data center electricity demand forecasts exhibit a different uncertainty structure, diverging much earlier and showing substantially higher implied growth intensity over shorter forecast horizons. Significant differences across institutional forecasts are already visible before 2030, both in target-year electricity demand levels and in CAGR-based growth rankings. This suggests that data center electricity demand uncertainty is not a smaller-scale version of national uncertainty, but a distinct forecasting challenge shaped by rapidly evolving AI adoption, computational demand growth, infrastructure deployment, server and cooling efficiency improvements and grid constraints.

Forecast uncertainty exists not only across institutions, but also within individual institutions. Sources such as EIA, NREL, LCRI, EPRI, BCG, and Rhodium express uncertainty through multiple scenarios with substantially different growth pathways. In some cases, within-institution scenario differences are comparable to or even larger than the differences between central forecasts from different institutions. This indicates that interpreting only a single representative forecast value is insufficient for understanding the uncertainty. For energy system planning, it is more important to understand the scenario boundaries, assumption ranges and structural uncertainty in each forecasting framework.

Overall, the results suggest that future U.S. electricity demand should not be interpreted as a single deterministic trajectory. Instead, it should be understood as a structured range of possible pathways

shaped by institutional assumptions, technological uncertainty, infrastructure constraints, and the evolving role of AI-driven data center expansion in the broader electricity system.

6.2 Uncertainty of Data Center Demand

Data center electricity demand is one of the most important, fastest-growing, and most uncertain load categories examined in this study. Compared with national electricity demand forecasts, data center-specific forecasts are characterized by shorter time horizons but earlier divergence. National electricity demand forecasts usually show more obvious structural divergence around 2040 or 2050, whereas data center electricity demand forecasts already show substantial differences around 2028 to 2030.

This near- to medium-term uncertainty is mainly driven by rapid changes in AI and cloud-computing infrastructure. Data center load growth is affected by multiple factors, including the rate of AI adoption, model training and inference demand, server deployment scale, chip efficiency, rack power density, server utilization, cooling energy consumption, grid interconnection capacity, and regional power infrastructure constraints. These variables are changing quickly and interact with one another, making it difficult to form stable and consistent long-term forecasts at the current stage.

The uncertainty of AI-related load is particularly important. Traditional data center demand has been driven mainly by cloud computing, enterprise IT services, internet applications, and data storage needs. In recent years, however, the new uncertainty has increasingly come from large-scale AI training and inference. Training loads may be concentrated in a limited number of large computing clusters, while inference loads may expand over time as AI applications become more widely adopted. Differences in institutional assumptions about the balance between training and inference demand, the speed of model efficiency improvements, hardware iteration, and application adoption can lead to large differences in data center electricity demand forecasts.

Moreover, data center electricity consumption depends not only on computing demand itself, but also on whether supporting infrastructure can be built in time. Even if market demand is strong, actual data center load growth may be lower than high-growth forecasts if grid interconnection, substation construction, transmission expansion, power supply, or land permitting become binding constraints. Conversely, if hyperscale companies, colocation providers, and utilities can coordinate power infrastructure development quickly, data center load growth may move closer to high-growth scenarios. Therefore, uncertainty in data center forecasts includes both demand-side uncertainty and supply-side interconnection constraints.

The results in Chapter 5 show that institutional views of U.S. data center electricity demand in 2030 already differ substantially. This indicates that data center forecasting is not a problem that can wait for long-term observation before being addressed; it already creates immediate planning pressure. For utilities, grid operators, and regulators, 2030 is not a distant year but a key point within current investment and resource planning cycles. Therefore, uncertainty in data center electricity demand should be explicitly incorporated into near-term planning rather than treated only as a long-term scenario issue.

6.3 Data Center Load Growth in the National Electricity System

Although data center electricity demand remains smaller than total U.S. electricity consumption in absolute terms, its system impact cannot be evaluated only by its share of national electricity use. Data center loads have several features that distinguish them from ordinary loads: rapid growth,

geographic concentration, large project-level capacity, high load persistence, high reliability requirements, and the need for large-scale power interconnection within relatively short timeframes. These characteristics mean that the impact of data center load on the power system may exceed what its national TWh share alone would suggest.

First, data center load is highly concentrated. Total national electricity demand is distributed across the residential, commercial, industrial, transportation, and other sectors, whereas large data centers are often concentrated in a limited number of regions or grid nodes. Even if a region accounts for only a small share of national electricity consumption, it may face local grid stress if multiple large data center projects are added within a short period. Therefore, data center load planning is not only a national energy demand issue, but also a regional grid planning and interconnection planning issue.

Second, data centers usually have high load factors. Many residential and commercial loads exhibit significant daily and seasonal variation, whereas data centers typically operate continuously. This means that the same interconnection capacity may correspond to higher annual electricity consumption and more persistent grid pressure. For generation resource planning and capacity planning, data center load not only increases annual electricity demand, but may also affect system peak demand, reserve requirements, and resource adequacy.

Third, data center load growth may change the pace of utility investment. Traditional electricity demand forecasts were relatively flat for a long period in many regions, so some grid planning processes may have been based on low-growth assumptions. However, rapid expansion of AI and data centers may push certain regions back into a high-load-growth environment. This can affect the planning pace for transmission lines, distribution systems, substations, generation capacity, long-term power purchase agreements, and clean electricity procurement.

Fourth, data center load may also affect low-carbon transition goals. If incremental data center electricity demand is met primarily by marginal fossil generation, it could increase emissions pressure in the power system. If it is supplied through renewable energy, nuclear power, storage, demand response, or clean electricity procurement, it could instead stimulate investment in low-carbon resources. Differences in clean electricity availability, temporal matching, and regional matching requirements across data center operators and electricity markets will also affect the actual emissions impact of data center growth.

Therefore, the importance of data center load in the broader power system is not simply a matter of its share of national electricity consumption. The more important questions are where these loads grow, when they connect to the grid, at what capacity scale they connect, what resources supply them, and whether they align with grid expansion and low-carbon transition goals. This is why this study analyzes national electricity demand and data center electricity demand separately, while ultimately placing them within the same research framework.

6.4 Implications for Energy Planning

The findings of this study have direct implications for energy planning.

First, future electricity demand planning should not rely on a single baseline forecast. Since substantial differences exist both across institutions and across scenarios within institutions, policymakers and energy planners may underestimate demand-growth uncertainty if relying on a single forecast.

Second, planning should use a forecast envelope rather than a single value. It helps planners identify plausible demand ranges, upper and lower bounds and key sources of uncertainty. Planning processes

should preserve sufficient capacity margin and flexibility to address possible load growth beyond current expectations and to support system reliability and operational performance.

Third, data center load should be incorporated into near-term planning sensitivities. According to the results, uncertainty in data center electricity demand is already significant. Therefore, utilities and grid planners should include different data center growth cases in near-term planning, and evaluate their effects on regional peak demand, transmission congestion, generation resource needs, and interconnection queues.

Fourth, power infrastructure planning needs to pay greater attention to spatial dimensions. National forecasts can describe macro-level trends, but they cannot directly determine whether specific regions have sufficient grid capacity to support data center interconnection. This is especially important in the United States, where the power system is divided among multiple regional grid operators and utility territories rather than a nationally unified grid. In data center cluster regions, local grid constraints may become binding earlier than national demand totals would suggest.

Finally, policy design needs to balance economic development, energy security, and low-carbon transition. AI and data center investment may bring economic growth, employment and technological innovation, but it can also increase pressure on the power system and the environment. Without forward-looking planning, this may lead to grid interconnection delays, electricity price pressure, increased fossil generation, or challenges to clean energy targets.

6.5 Limitations

This study has several limitations.

First, the data sources are highly heterogeneous. Although this study uses a standardized database to improve comparability, these sources are not generated within the same modeling framework. In addition, many studies do not fully disclose detailed assumptions or complete forecast datasets, which limits the possibility of conducting a fully controlled variable-by-variable comparison.

Second, some data were obtained through image digitization or derived calculations, which introduces potential error. Not all institutions provide complete annual data tables. Some sources provide only graphical curves, benchmark-year values, growth rates or share-based information. Although this study preserves original source information and records processing methods, digitization and derivation inevitably introduce some uncertainty.

Third, scenario standardization involves a degree of subjectivity. This study classifies original scenarios from different sources into low, mid and high categories to support database management and cross-source comparison. However, these three categories do not necessarily have the same probabilistic or policy meaning across institutions. They should be treated as comparison tools rather than a way to eliminate differences in original scenario definitions.

Fourth, data coverage is incomplete for some years and sources. National electricity demand forecasts often extend to 2050, while data center-specific forecasts mostly end around 2030. This limits the ability to systematically compare long-term data center electricity demand trends. The short horizon of data center forecasts is nonetheless understandable, because data center demand is changing so quickly that long-horizon forecasts would carry very large uncertainty.

Fifth, this study mainly focuses on annual demand rather than hourly load, peak demand, capacity requirements or regional congestion. For power system planning, annual demand is important, but it does not fully capture operational stress. This is especially important for data center load because its

high load factor and concentrated interconnection characteristics may have important effects on peak demand, local transmission constraints, and resource adequacy.

6.6 Future Work

Future research can extend this study in several directions.

To begin with, the database can be continuously updated. AI technologies, data center construction, and energy policies are changing quickly, so existing forecasts may be revised substantially when new data and assumptions become available. Electricity demand forecasts should be treated as a dynamic database that requires continuous updating.

Then, the analysis can be expanded to finer spatial scales. Future research can include state-level, ISO/RTO-level, utility territory, or data center cluster-level data to examine how data center load growth affects regional power systems. National-level analysis can describe macro-level trends, but regional analysis is more useful for supporting specific grid planning and investment decisions.

In addition, future research can further analyze the share of data center electricity demand in national demand. Although this study collects some share-related data, complete and comparable share data remain limited. If more paired forecasts for the same years become available, such work could analyze more systematically how the data center expansion shapes the U.S. energy system over time.

Moreover, future work can incorporate hourly load profiles and peak demand analysis. The system impact of data center load depends not only on annual electricity consumption, but also on load shape, peak contribution, coincidence factor, and operational flexibility. Future research can convert annual TWh forecasts into load-curve scenarios and further analyze their effects on generation capacity, storage requirements, transmission congestion, and system reliability.

Furthermore, future research can conduct scenario sensitivity analysis. A sensitivity framework can be built around key variables such as AI adoption rate, GPU efficiency, server utilization, cooling efficiency, power usage effectiveness (PUE), chip power density, renewable procurement, and grid interconnection delays. This would make it possible to identify more clearly which factors have the largest effects on data center electricity demand forecasts.

Finally, electricity demand forecasts can be linked with emissions and energy mix analysis. Data center growth may be supplied by different generation resources, and its impact on carbon emissions and low-carbon transition goals depends on the marginal generation mix, clean electricity procurement, and regional electricity market rules. Future research can further evaluate generation structure, emissions changes, and policy implications under different data center growth scenarios.

Overall, future work can build on the standardized database and comparison framework developed in this study and expand it toward spatially detailed, temporally resolved, dynamically updated, and system-planning-oriented analysis. The findings of this study show that U.S. electricity demand forecasting in the AI era is no longer only a traditional macro-level demand forecasting problem, but a broader energy-system issue involving data center load growth, grid interconnection, low-carbon transition, and regional planning.

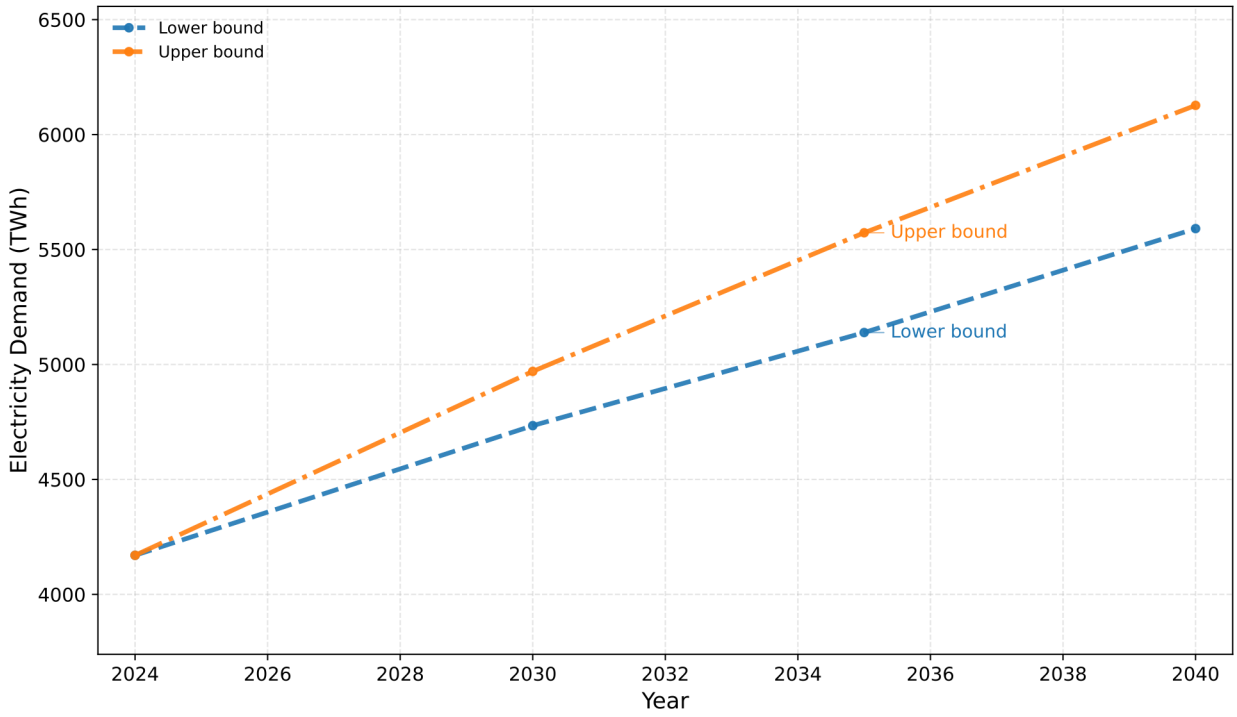
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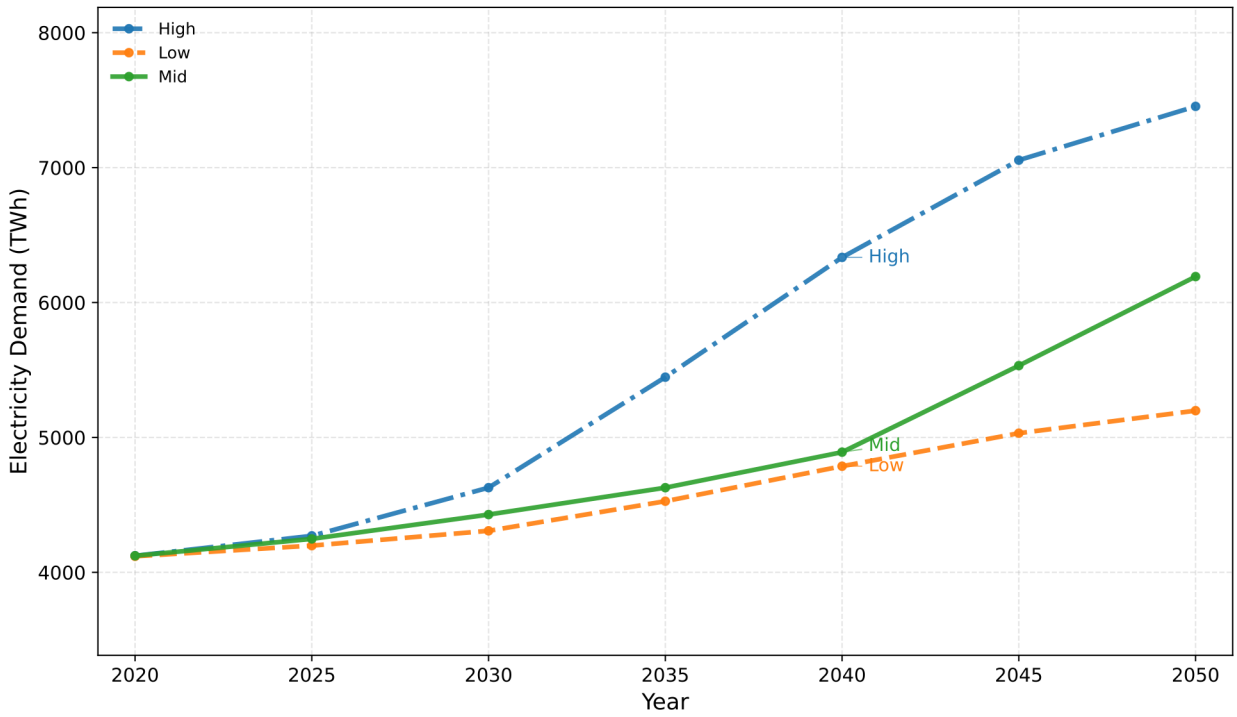
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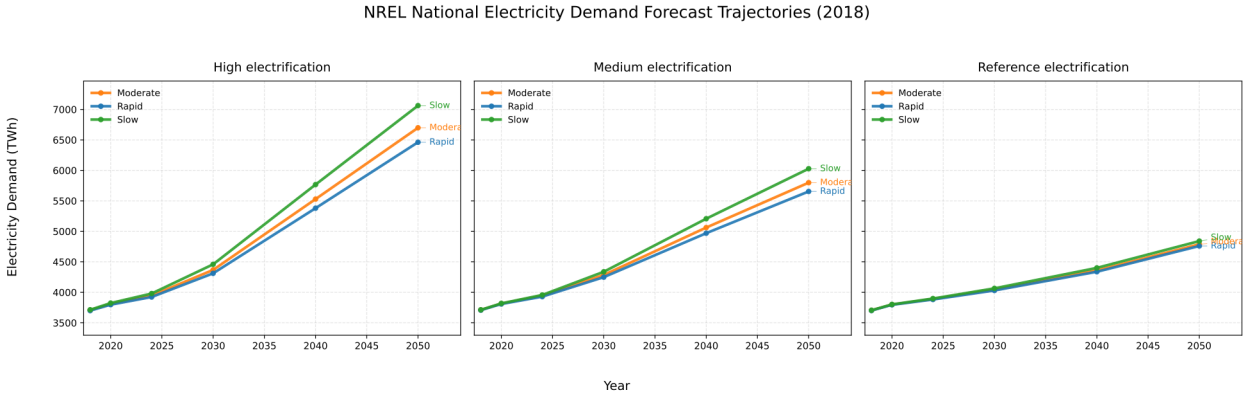
Appendix I: Single-Institution National Electricity Demand Forecasts

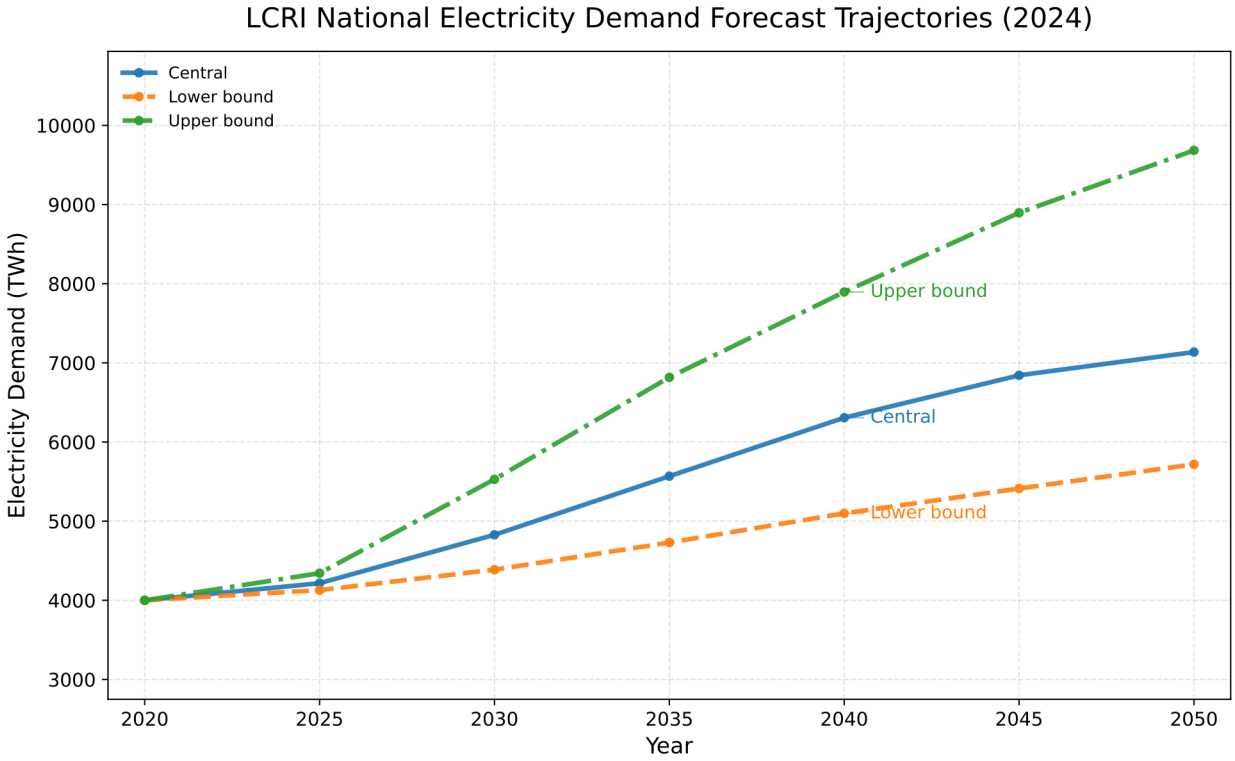
S&P National Electricity Demand Forecast Trajectories (2025)

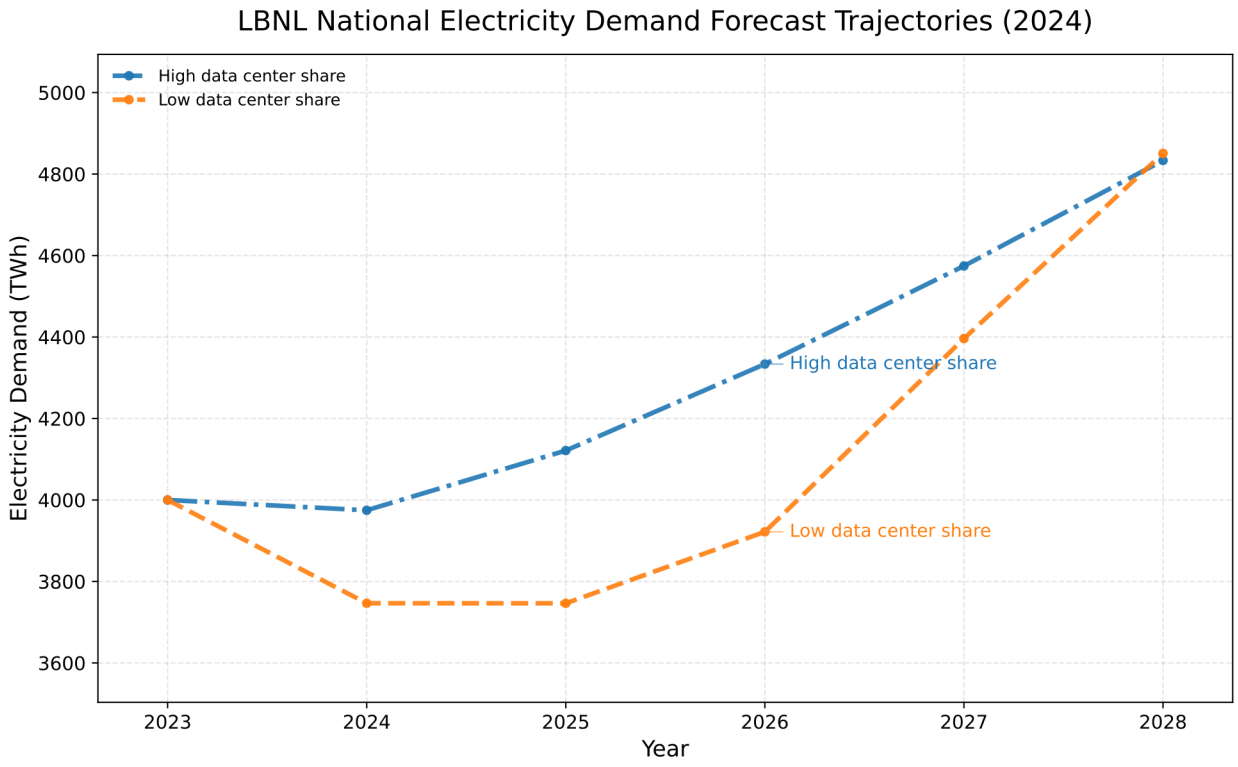


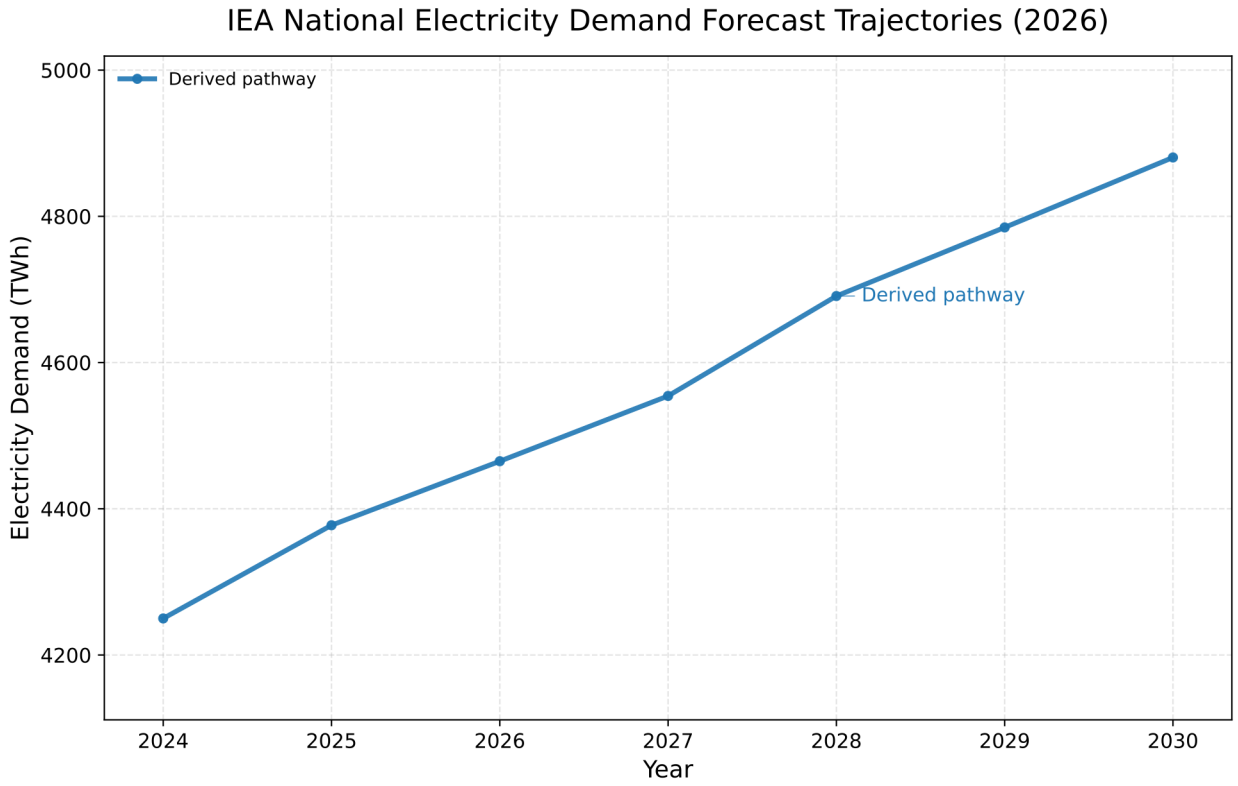
Princeton National Electricity Demand Forecast Trajectories (2021)

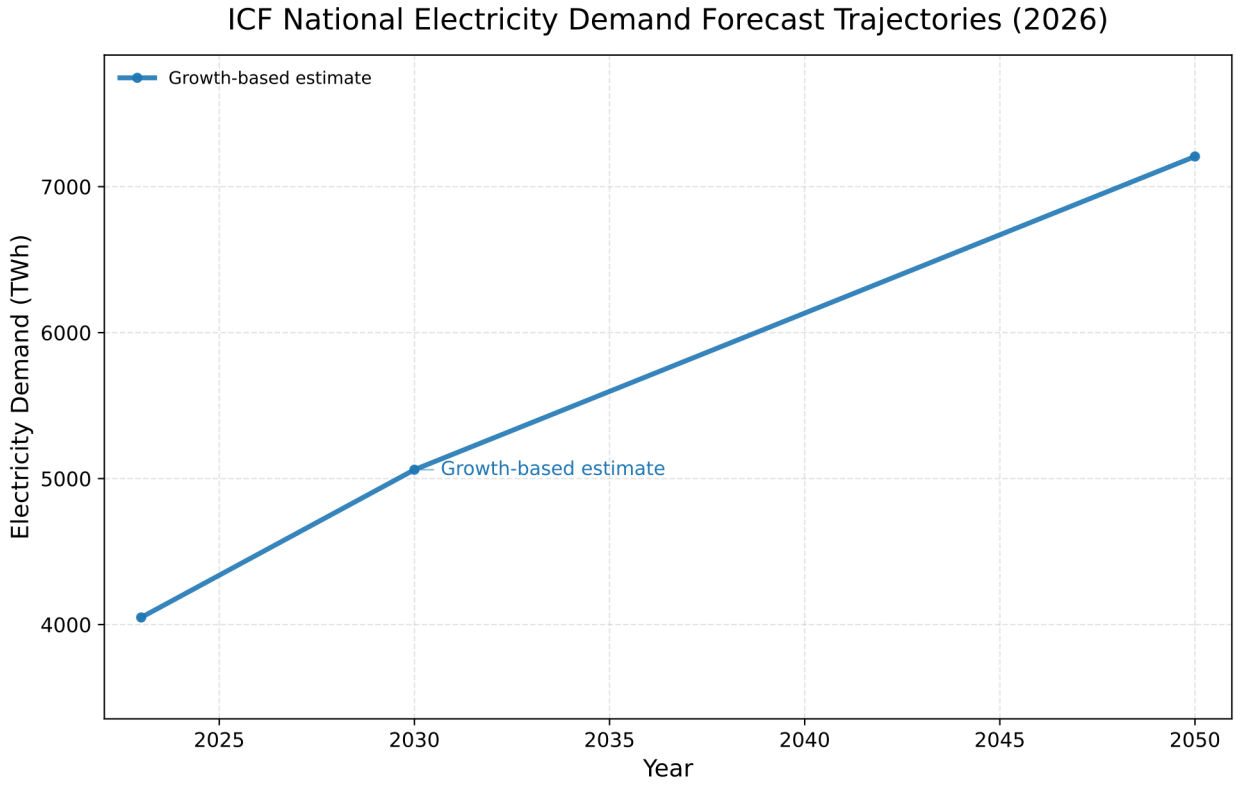


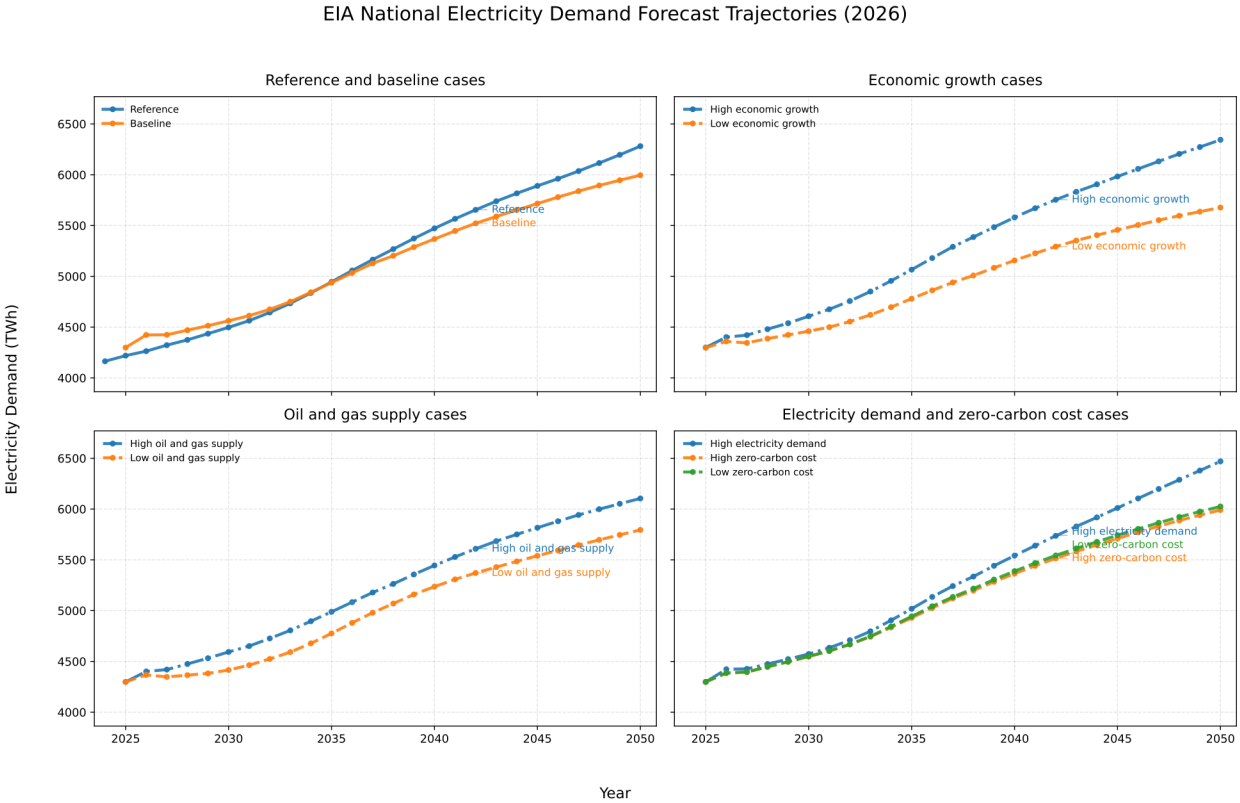

NREL National Electricity Demand Forecast Trajectories (2018)
High electrification
Medium electrification
Reference electrification
Moderate
Rapid
Slow
Electricity Demand (TWh)
Year

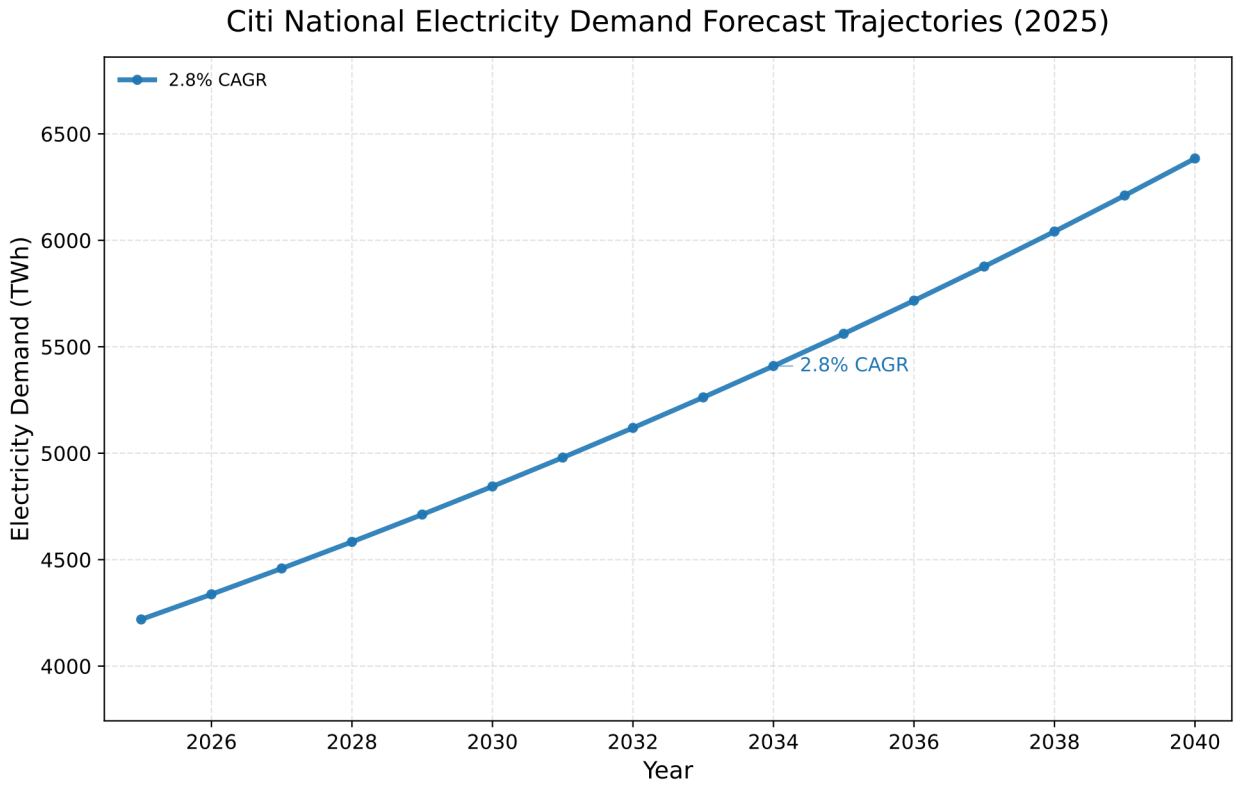

LCRI National Electricity Demand Forecast Trajectories (2024)
Central
Lower bound
Upper bound
Electricity Demand (TWh)
Year

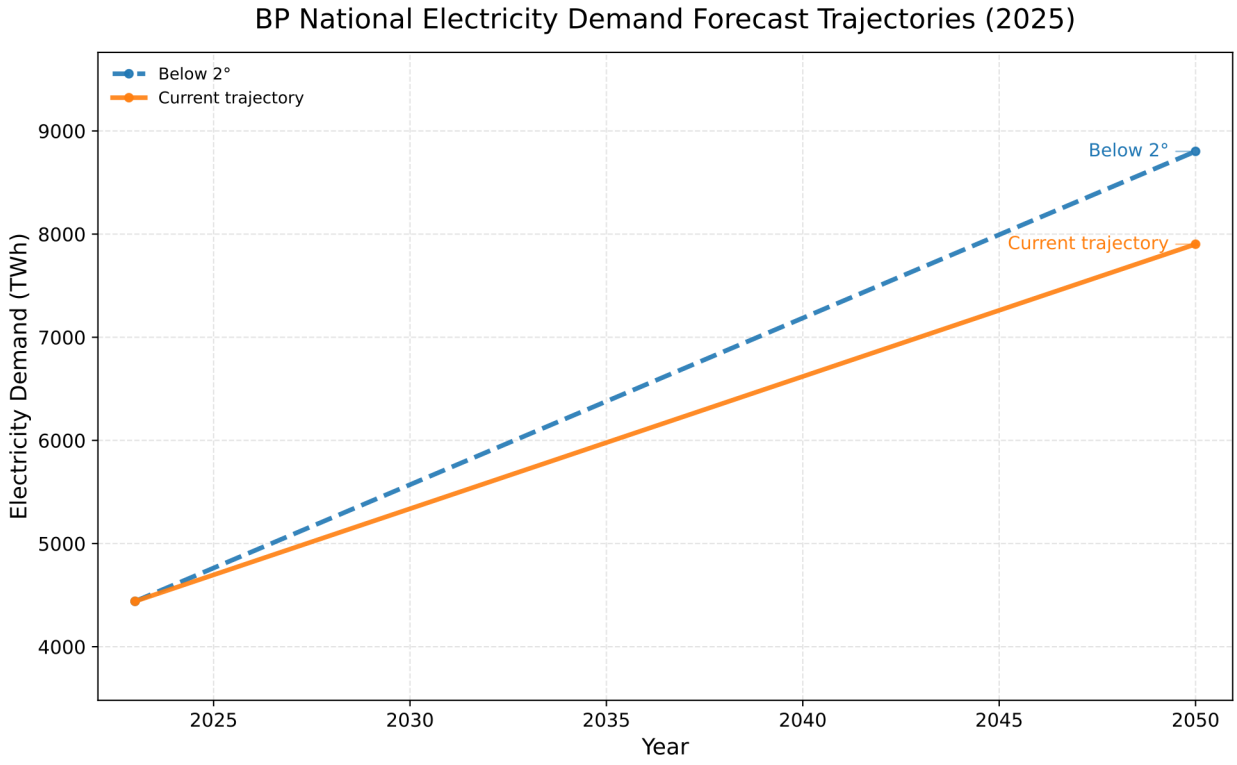

LBNL National Electricity Demand Forecast Trajectories (2024)
High data center share
Low data center share
Electricity Demand (TWh)
5000
4800
4600
4400
4200
4000
3800
3600
2023
2024
2025
2026
2027
2028
Year
High data center share
Low data center share


IEA National Electricity Demand Forecast Trajectories (2026)
Derived pathway
Electricity Demand (TWh)
5000
4800
4600
4400
4200
2024
2025
2026
2027
2028
2029
2030
Year
Derived pathway


ICF National Electricity Demand Forecast Trajectories (2026)
Growth-based estimate
Electricity Demand (TWh)
7000
6000
5000
4000
Growth-based estimate
2025
2030
2035
2040
2045
2050
Year

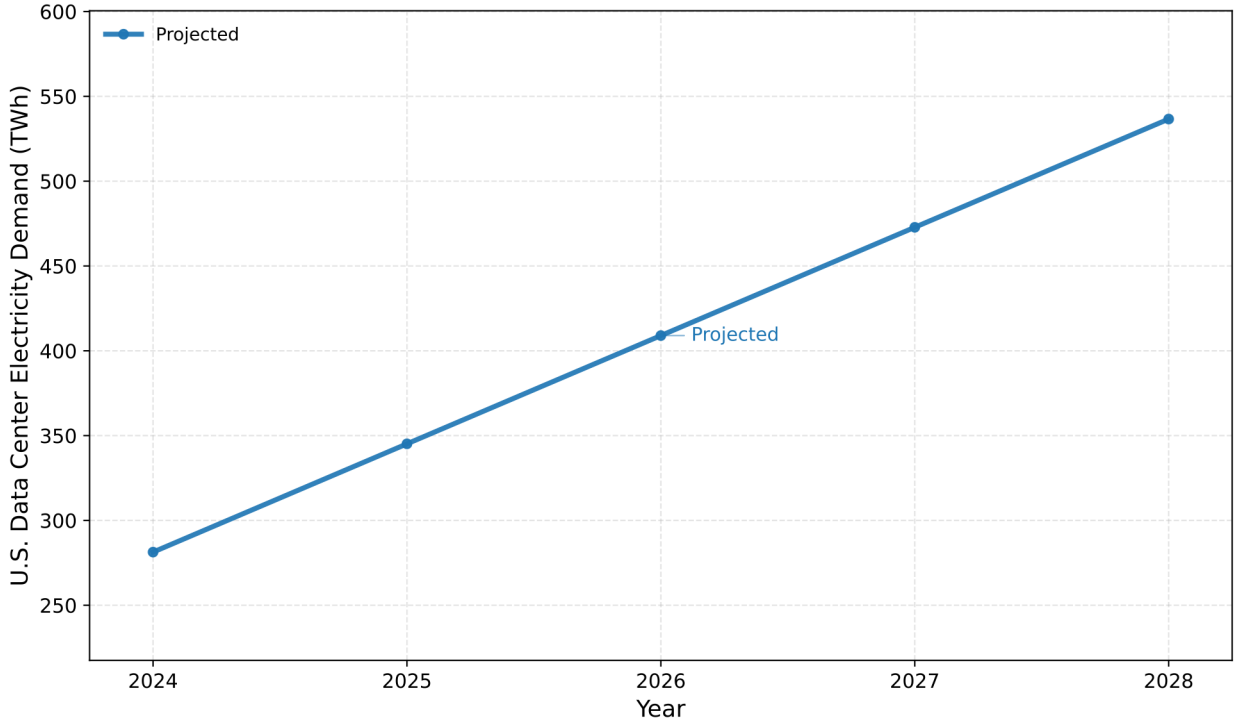

EIA National Electricity Demand Forecast Trajectories (2026)
Reference and baseline cases
Reference
Baseline
Economic growth cases
High economic growth
Low economic growth
Oil and gas supply cases
High oil and gas supply
Low oil and gas supply
Electricity demand and zero-carbon cost cases
High electricity demand
High zero-carbon cost
Low zero-carbon cost
Electricity Demand (TWh)
Year


Citi National Electricity Demand Forecast Trajectories (2025)
2.8% CAGR
2.8% CAGR
Electricity Demand (TWh)
Year
4000
4500
5000
5500
6000
6500
2026
2028
2030
2032
2034
2036
2038
2040

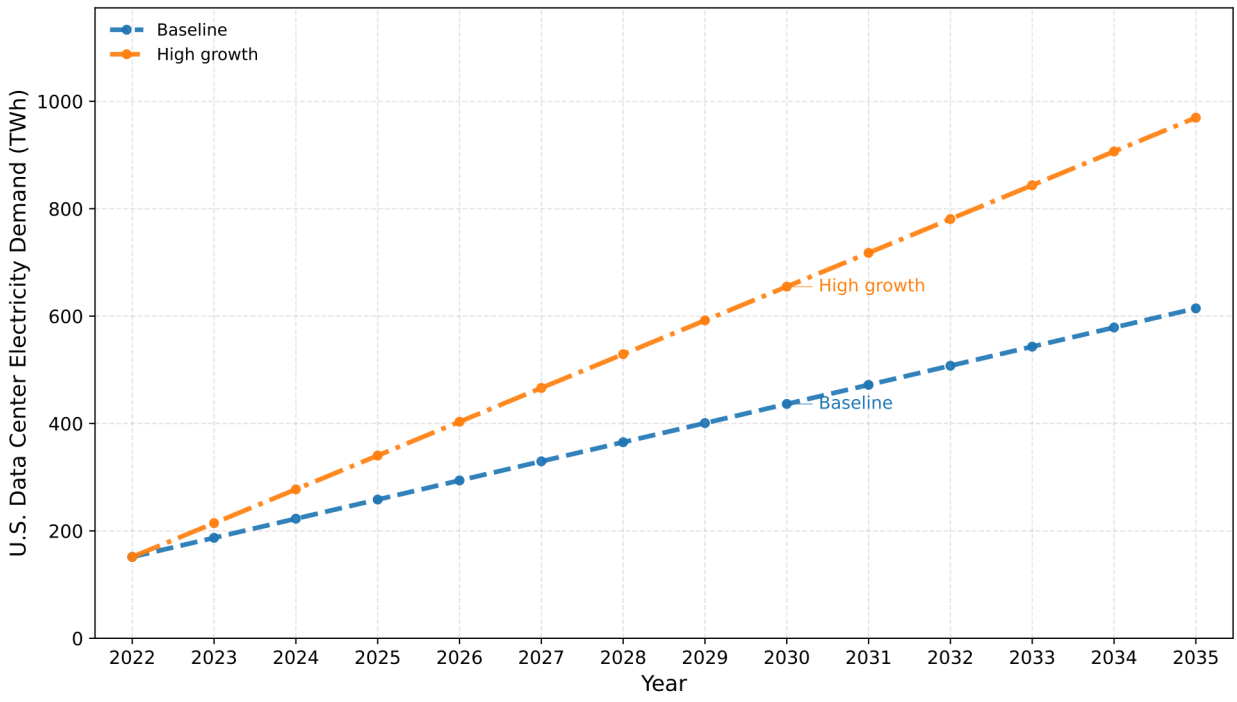

BP National Electricity Demand Forecast Trajectories (2025)
Below 2°
Current trajectory
Below 2°
Current trajectory
Electricity Demand (TWh)
Year
4000
5000
6000
7000
8000
9000
2025
2030
2035
2040
2045
2050

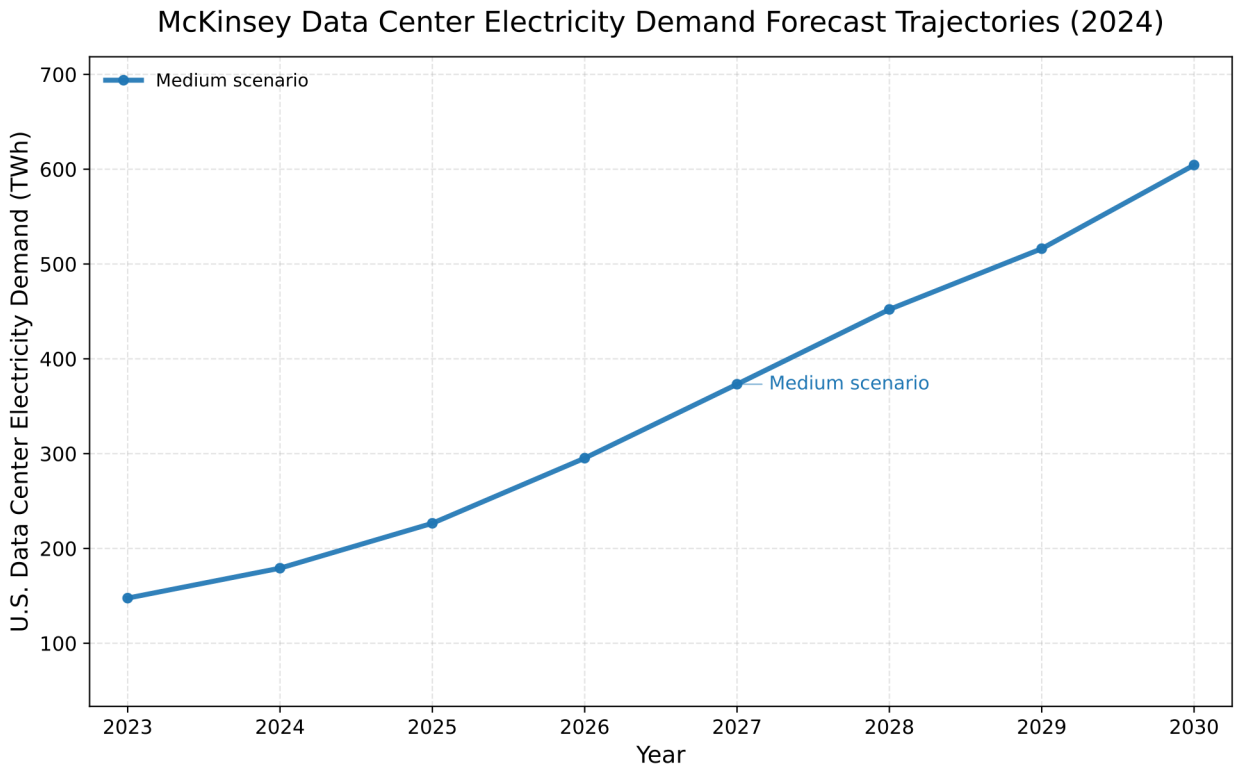
Appendix II: Single-Institution Data Center Electricity Demand Forecasts

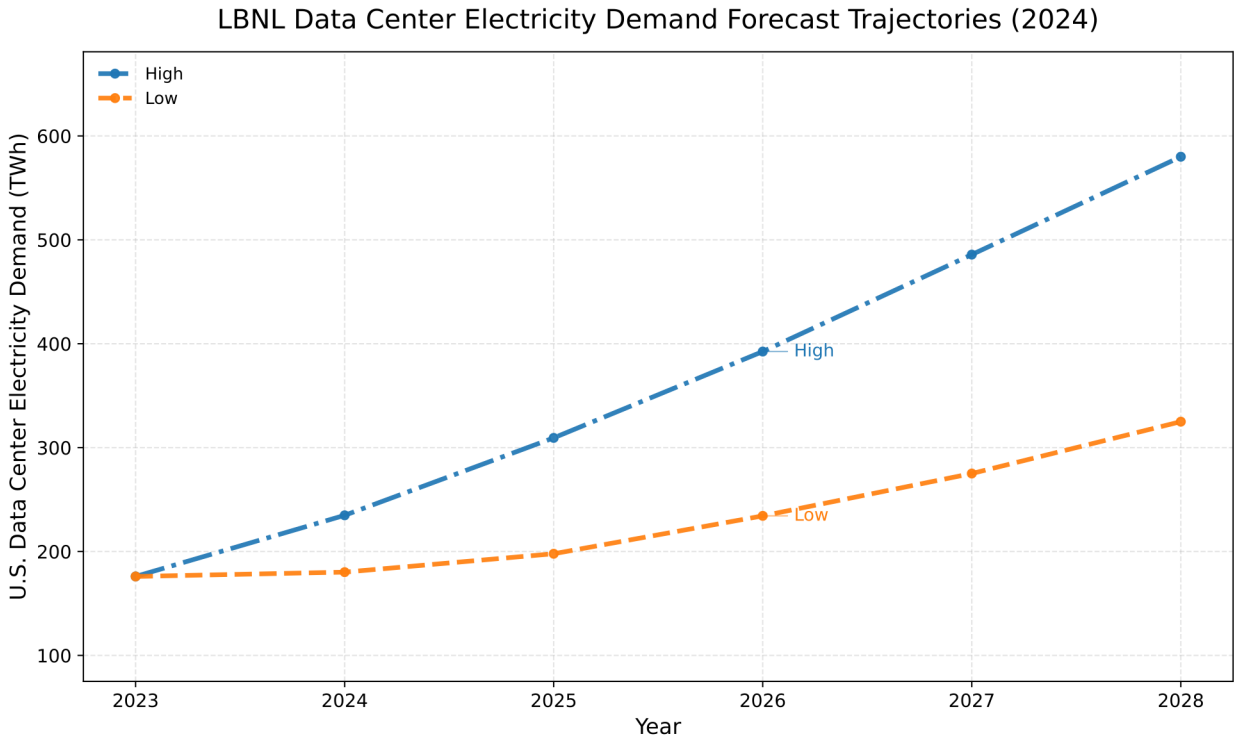
S&P Data Center Electricity Demand Forecast Trajectories (2024)

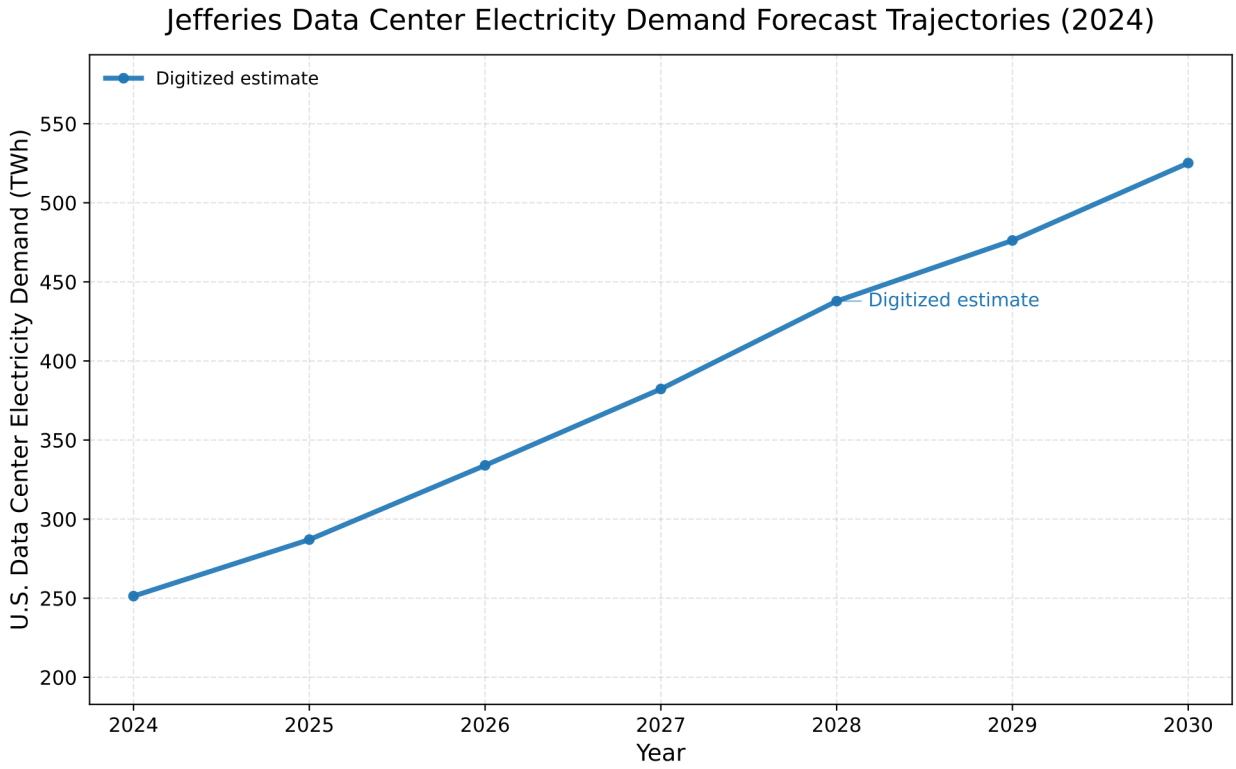


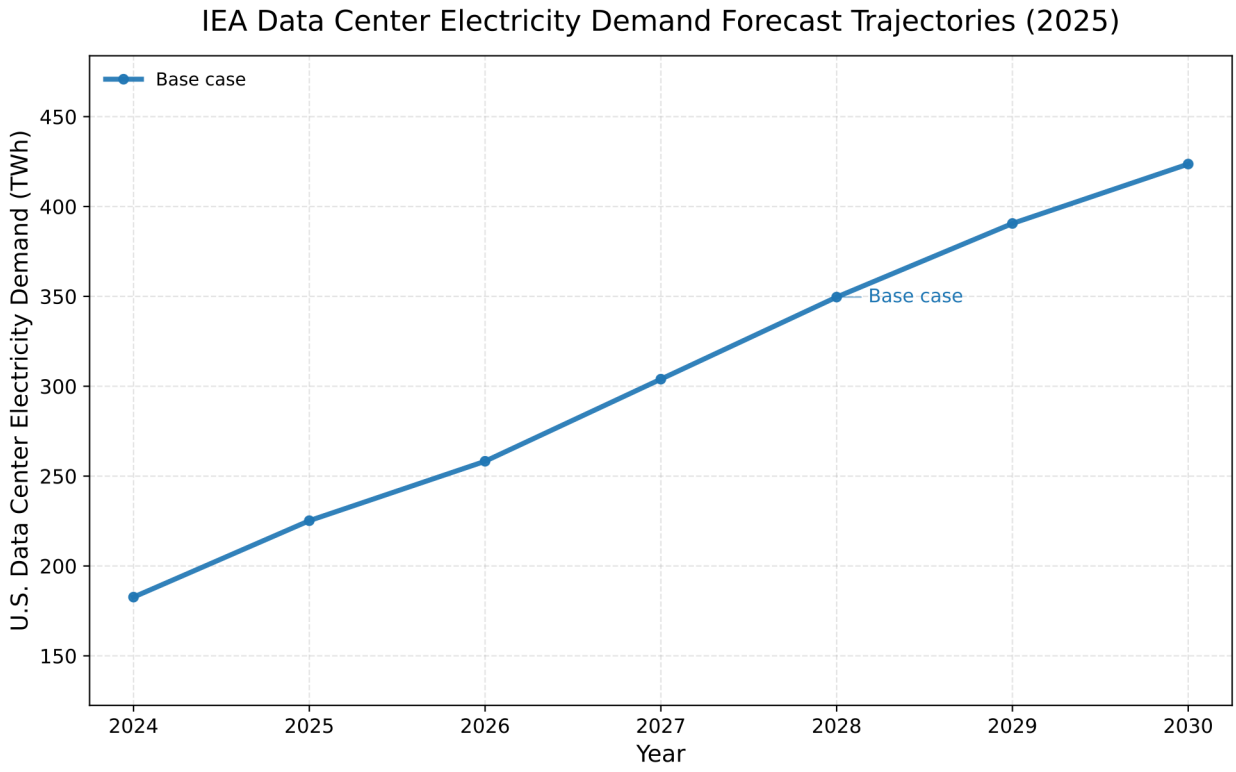
Rhodium Data Center Electricity Demand Forecast Trajectories (2026)

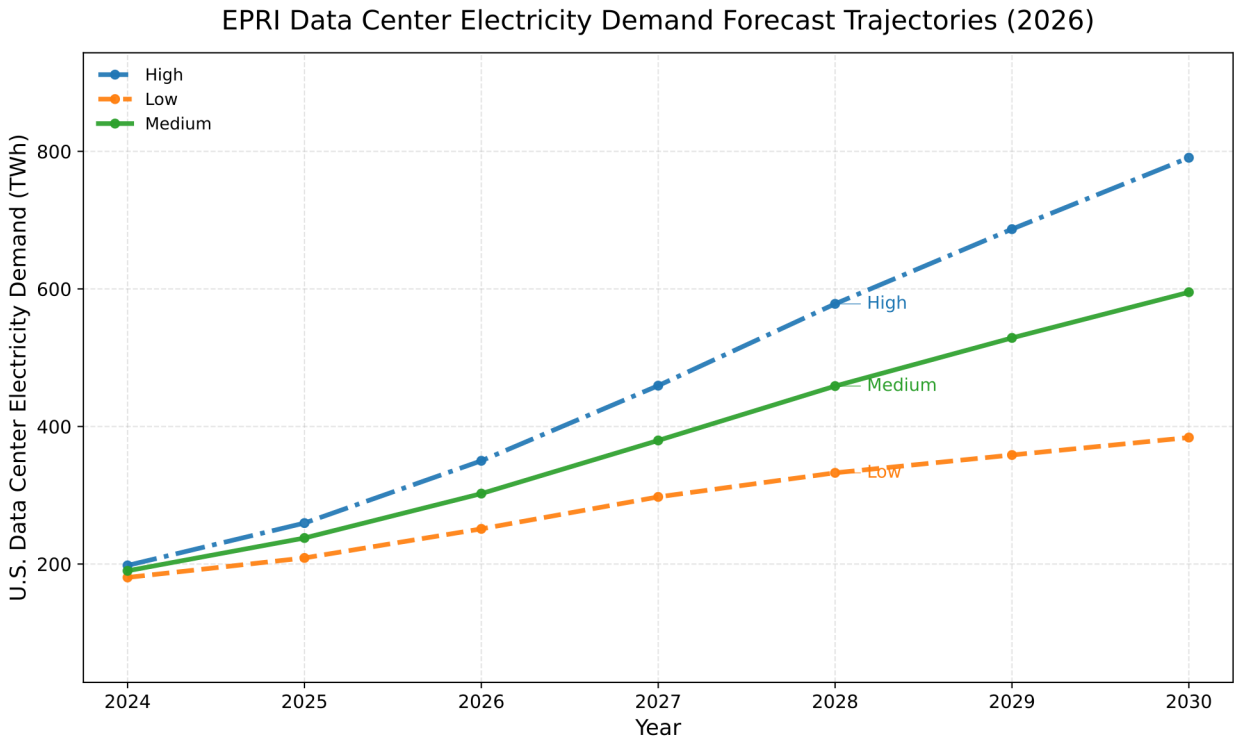


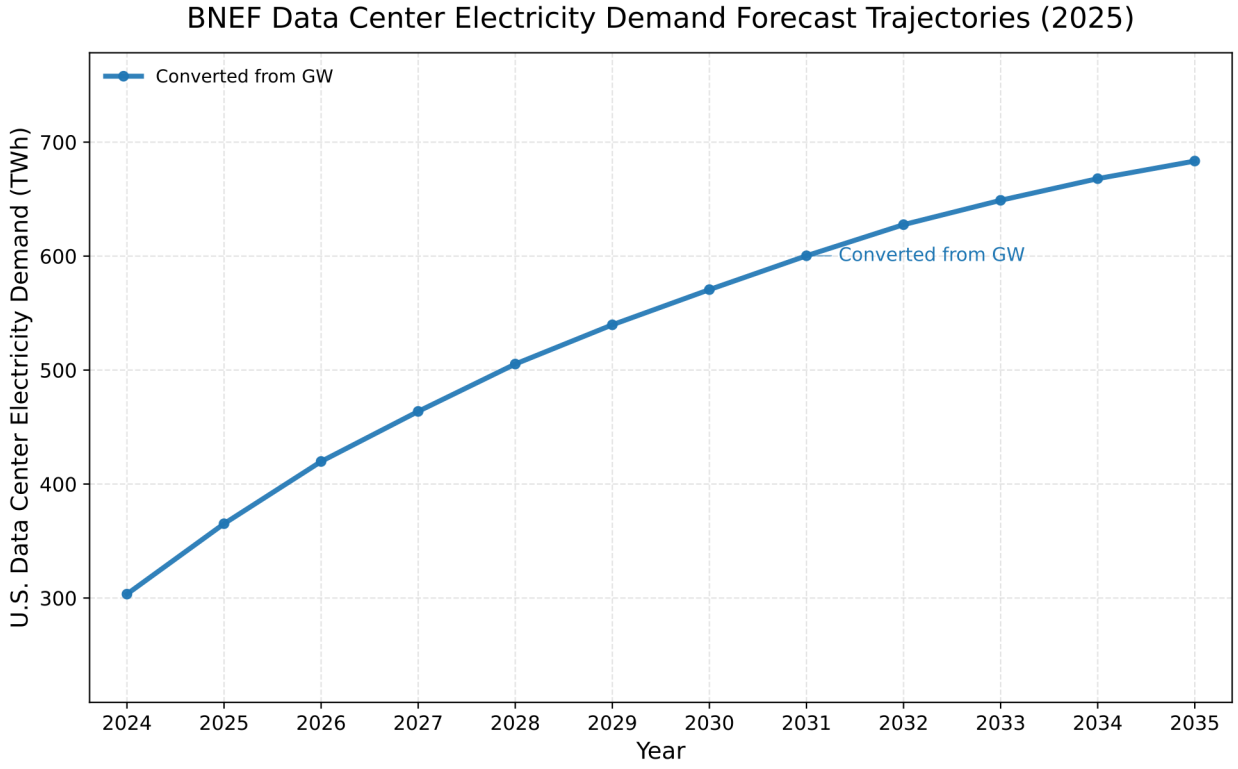

McKinsey Data Center Electricity Demand Forecast Trajectories (2024)
Medium scenario
U.S. Data Center Electricity Demand (TWh)
700
600
500
400
300
200
100
Medium scenario
2023
2024
2025
2026
2027
2028
2029
2030
Year

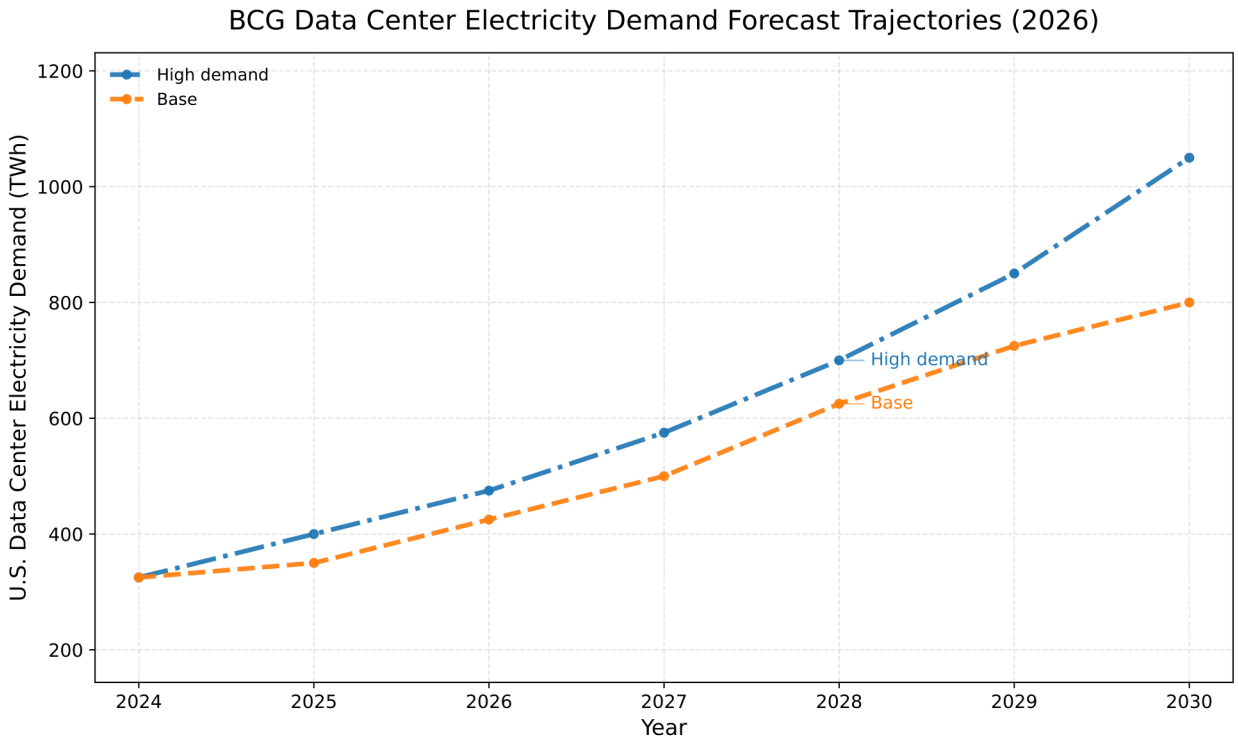

LBNL Data Center Electricity Demand Forecast Trajectories (2024)
High
Low
U.S. Data Center Electricity Demand (TWh)
600
500
400
300
200
100
High
Low
2023
2024
2025
2026
2027
2028
Year


Jefferies Data Center Electricity Demand Forecast Trajectories (2024)
Digitized estimate
U.S. Data Center Electricity Demand (TWh)
550
500
450
400
350
300
250
200
Digitized estimate
2024
2025
2026
2027
2028
2029
2030
Year


IEA Data Center Electricity Demand Forecast Trajectories (2025)
Base case
U.S. Data Center Electricity Demand (TWh)
450
400
350
300
250
200
150
Base case
2024
2025
2026
2027
2028
2029
2030
Year


EPRI Data Center Electricity Demand Forecast Trajectories (2026)
High
Low
Medium
U.S. Data Center Electricity Demand (TWh)
800
600
400
200
2024
2025
2026
2027
2028
2029
2030
Year
High
Medium
Low


BNEF Data Center Electricity Demand Forecast Trajectories (2025)
Converted from GW
U.S. Data Center Electricity Demand (TWh)
700
600
500
400
300
2024
2025
2026
2027
2028
2029
2030
2031
2032
2033
2034
2035
Year
Converted from GW


BCG Data Center Electricity Demand Forecast Trajectories (2026)
High demand
Base
U.S. Data Center Electricity Demand (TWh)
1200
1000
800
600
400
200
2024
2025
2026
2027
2028
2029
2030
Year
High demand
Base